

Umwelt- und Meerestechnik Kiel GmbH

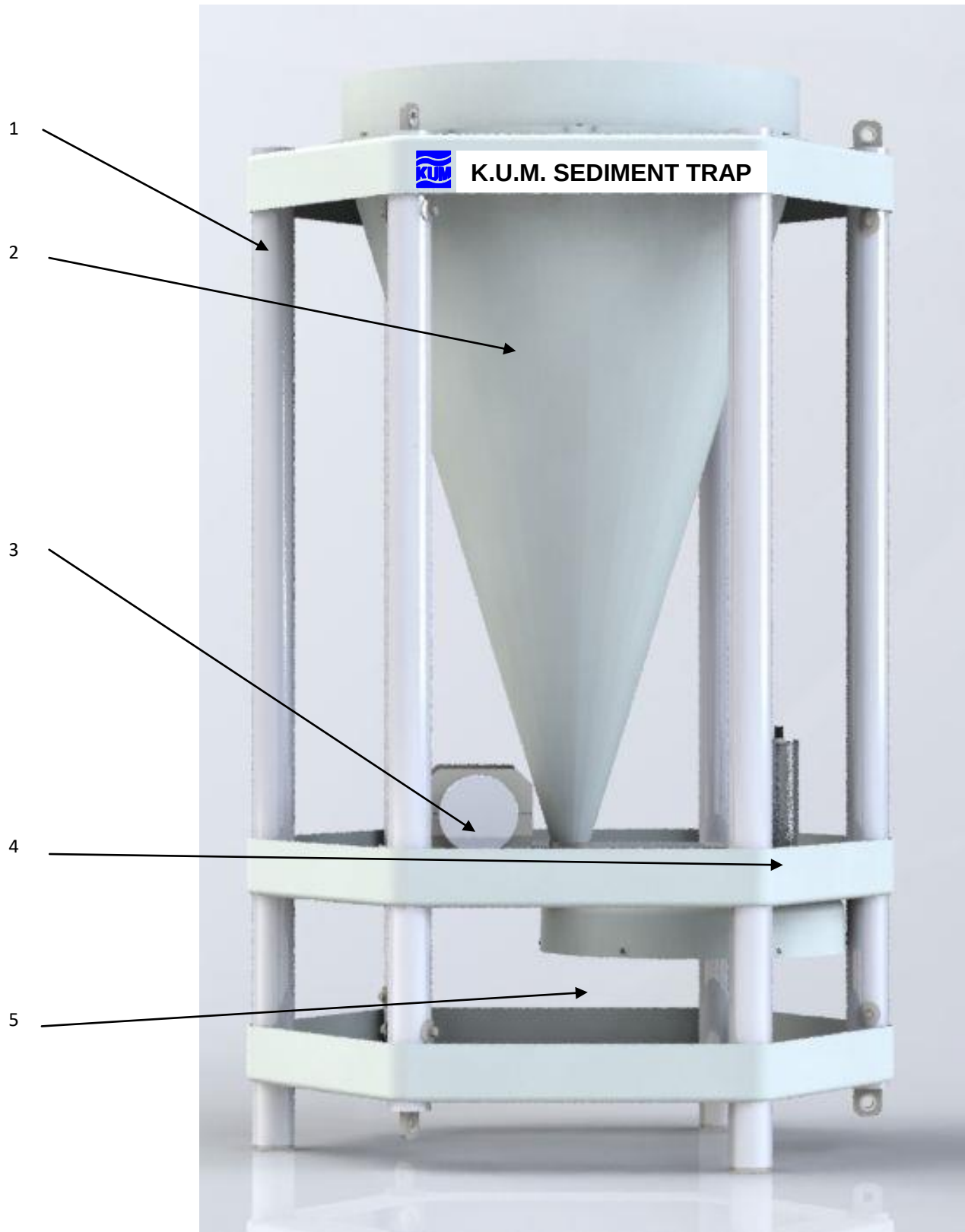
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S E D I M E N T T R A P

Manual 2019





- 1 Frame
- 2 Conus / Funnel with protection
- 3 Electronics and batteries
- 4 Drive Unit
- 5 Sample unit

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1. GENERAL INSTRUCTIONS

1.1 GENERAL SAFETY REGULATIONS

The legal 'Regulations for Prevention of Accidents' are decisive for maintenance and repair work.

Attention: Never turn the sampler outside the water! Severe damage might be the result!

The corresponding operating instructions must be observed for work with the lifting jack or crane.

The individual components must be marked prior to each repair.

In order to avoid corrosion, only use impeccable tools of bronze or special steel 1.4539 when changing the battery as well as during maintenance and repair work. In principle, only use new sealing and locking elements for the assembly of devices.

Do not assemble screw connections when without specified lubricants.

All maintenance and repair work may only be performed by personnel with the appropriate special training.

The system has to be checked for operational ability after every maintenance and repair work.

Instructions for Cables and Lines

When disconnecting plug connections, existing caps on plugs and sockets must be fitted in order to avoid humidity and foreign substances as well as damage to the contacts.

Flexible supply lines and connecting cables must not be submitted to undue tensile stress during operation and transportation of the devices. Special care has to be taken that the devices are not suspended by means of supply lines and that plugs are not unplugged from the sockets by pulling them out using the line.

Instructions for Plug Devices

Movable electrical devices (e.g. personal computer) may only be connected by means of the provided plug devices (plug, socket).

The application of adapters and plugs together with sockets of another voltage is inadmissible.

Plugs must not be drawn under load.

1.2 COMMENTS ON THE MANUAL

1.2.1 Preface

The present manual has been drawn up as a description and reference book. It will help you to answer questions and to solve problems in the quickest possible manner.

Please read the user's manual carefully if you have problems with your Sediment Trap.

For this purpose refer to the table of contents and read the corresponding section attentively.

If you have any further questions, please contact us under the following address stating the serial number :

1. email: kum@kum-kiel.de

2. by letter to:

K.U.M. Umwelt- und Meerestechnik Kiel GmbH
Wischhofstr. 1-3 Geb. 15
24148 Kiel
Germany

3. in very urgent cases by phone under:

Central office: Tel. +49(0)431/7209220

You should always have the manual at hand.

4. FAX: +49(0)431/7209244

2. TECHNICAL DESCRIPTION K/MT 234**2.1 TECHNICAL DATA**

Working depth:	max. 6000 m
Operating time:	24 months using fully charged batteries
Number of samples:	21
Contents of sample bottle:	400 ml (14 oz.)
Multi-sample changer:	microprocessor-controlled, interval between 2 samples is programmable between 1 hour and 24 months (clock, calendar)
Current supply:	12 V DC, using Mono type 1.5 V, alkaline LR20/AM1
Collecting surface:	0.5 m ²
Funnel angle:	34°
Funnel height:	1313 mm
Funnel diameter:	800 mm
Total height:	1900 mm
Width:	1079 mm
Material:	FRP sealed with special resin
Shape of casing:	hexagonal, employment of several traps in interconnected operation possible
Fastening:	a) one-sided suspension b) 3-point suspension for crowfoot (option)
Admissible temperature ranges:	
Operation:	-10 °C up to +40 °C
Storage and transport:	-30 °C up to +70 °C

2.2

SHORT DESCRIPTION OF DEVICE

The Sediment Trap serves to investigate the vertical particle flow in a water column (as a function of place and time) and can be used both in the ocean and in inland waters.

The Sediment Trap is moored at a specified water depth and stores the sinking down substances in a sample bottle. The substances sinking down above the Sediment Trap are collected by a funnel and collected into a sample bottle underneath it. After a pre-programmed time interval the next bottle is pushed under the funnel, ending the current sample and starting the next.

In a processor-controlled sample changer, up to 20 sample bottles can be installed for K/MT 234. The sample time of the individual sample bottles below the funnel is programmable between 1 hour and 24 months. A special construction is guaranteeing that those sample bottles not under the funnel are sealed against the outer medium.

The open frame construction keeps the flow resistance as low as possible and serves to receive the funnel and the sample changer. The special hexagonal shape allows the employment of several traps in interconnected operation. The material is a high-tensile FRP with specially sealed surface.

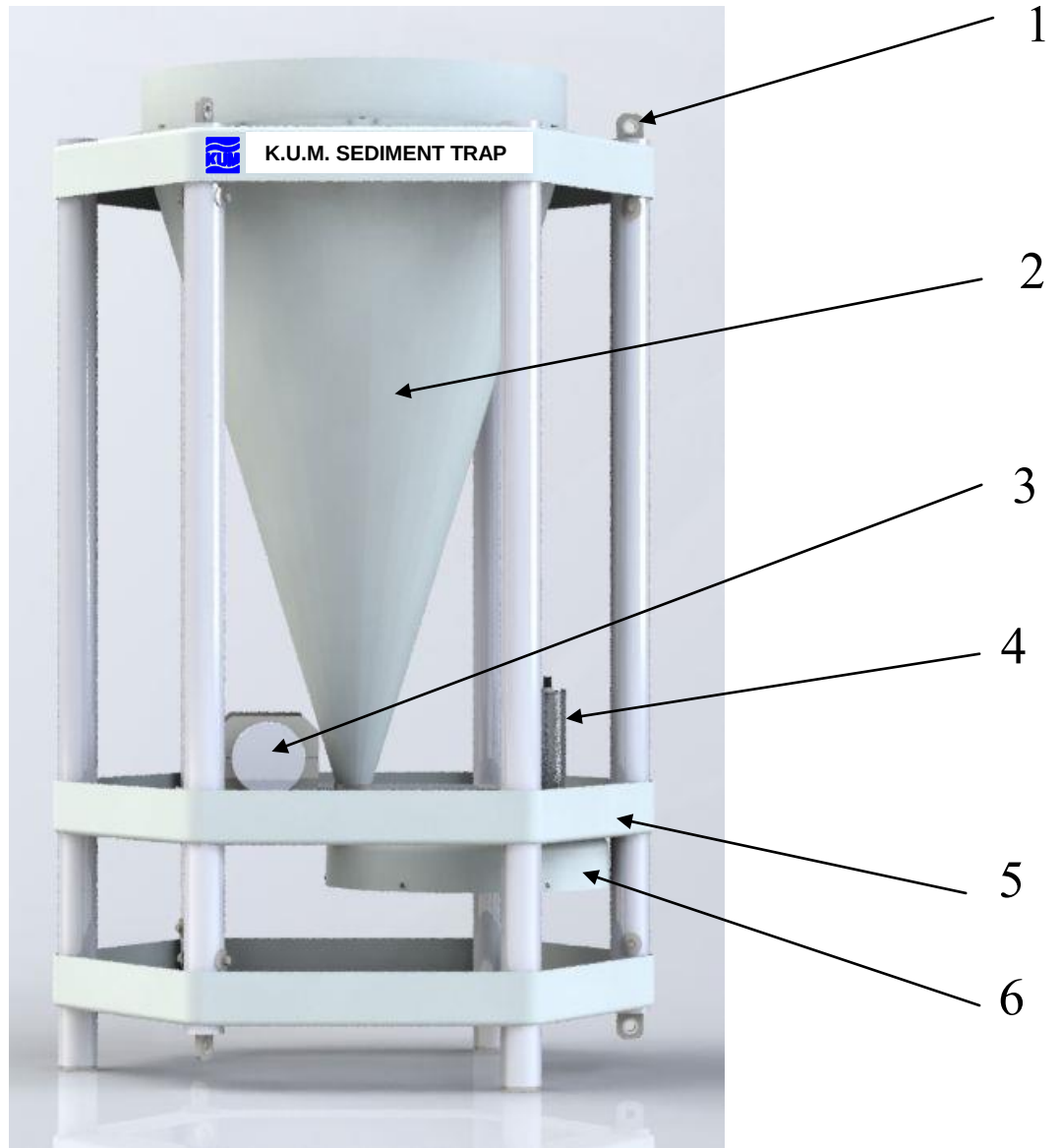
The funnel fixed to the frame construction collect the sediments. The sediments passed through the grating are collected and passed through the funnel to the sample bottles below the funnel. A hexagonal honeycomb-network of sealed special paper serves as funnel cover and to prevent flushes. Furthermore, it prevents the entry of larger particles that could block the opening of the sample bottle.

The sample changer consists of a base plate, below which the sample bottles are circularly arranged on a traveller ring. A motor positions the various sample bottles below the exit of the funnel. The sample changer is made from seawater-resistant plastic.

Programmable microprocessor electronics control the turntable. Programming is menu-guided via a laptop. The Sediment Trap connects to any standard WiFi device. After programming has been performed, the Sediment Trap is put into sleeping mode. Thus, even on-shore programming is possible. After recovery data can be saved and printed.

The control unit contains the batteries and is mechanically separated from the drive. They are connected to each other via a pressure-watertight plug connection. The control unit is fixed above the multi-sample changer in horizontal position. This guarantees an easy mounting and dismounting for programming, exchanging the battery and maintenance.

The control unit is equipped with integrated inclinometer. In case the Sediment Trap is tilted because of heavy current the effective collecting surface can be calculated to eliminate the error using the tilt recordings.



- | | |
|---|-----------------------|
| 1 One-sided fastening (reinforced spar) | 4 Drive unit |
| 2 Funnel | 5 Intermediate bottom |
| 3 Control unit | 6 Sample changer |

Fig 1: Set up of Sediment Trap

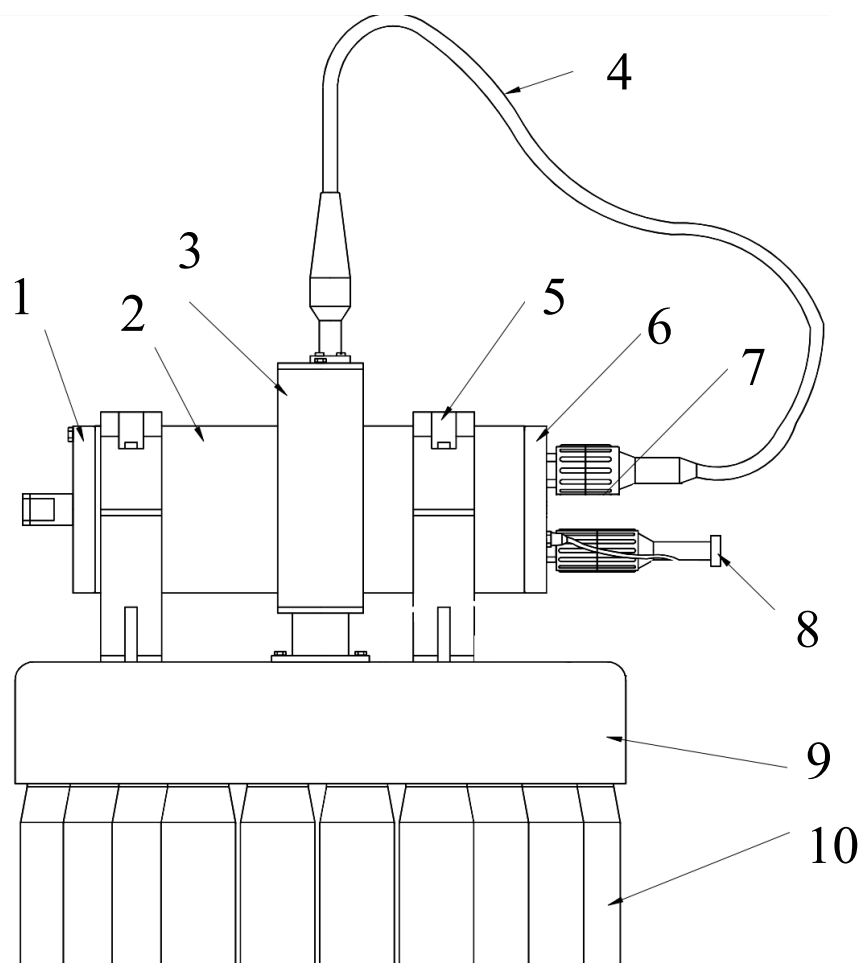
ATTENTION: For one-sided suspension of the Trap, the specially marked “MOORING SPAR” (1) must be used.

2.3.2 Structure of the Sample Changer

The structure of the sample changer is shown in the figure below.

The sample changer is composed of the turntable (9), the control unit (2) and the drive unit (3). The drive unit is electrically connected to the control unit by a pressure-watertight cable (4). To program the unit attach the WiFi antenna (8) to the control unit. The WiFi will be available after one minute. After programming it is mandatory to remove the WiFi antenna.

The control unit and the drive unit are mounted from above. Power is supplied by alkaline batteries within the control unit.



- | | |
|--|--|
| 1 Battery cover | 6 Bottom for control unit |
| 2 Control unit | 7 Connector to drive unit |
| 3 Drive unit | 8 Dummy, to be replaced by WiFi connector during programming |
| 4 Connecting cable to the control unit | 9 Turntable |
| 5 Fastener for control unit | 10 Sample bottle |

Figure 2: Structure of a sample changer

A hand-operated revolving mount (option) can be mounted onto the turntable in order to manually operate it. This makes it easier to install the sample bottles. For this purpose the drive unit is released from its support and removed. The hand-operated revolving mount is mounted in the place of the drive unit. The turntable can now be turned freely using a cam-bolt wrench or a ring-nut wrench of 13 mm width. The ring gear can also be forwarded using the webpage.

ATTENTION: Never turn the samples outside the water! Severe damage might be the result!

The sampling positions are indicated by numbers on the ring gear. They are visible from the interior side through an opening in the upper mounting plate below the funnel outlet. The sample bottles are marked on the field of the sample bottle with a watertight pencil.

The sample bottles are inserted from below into the bottle holders of the turntable. On the top side of the turntable there is an opening in order to fill sample bottles with fixing agent.

2.4 TECHNICAL DESCRIPTION OF ELECTRONICS

The control unit of the Sediment Trap has to perform two essential tasks:

- Time keeping and sensor data sampling
- Control of the sample change

After connecting with WiFi the required times for the 20 samples can be defined. After programming of the Sediment Trap electronics you ***must*** remove the WiFi antenna. The Sediment Trap will start its program. ***Make sure the Trap is submerged in water when the first sample is turned.***

3. ESTABLISHING THE OPERATING CONDITIONS

3.1 ELECTRIC CONNECTION

The drive unit is connected to the control unit by a pressure-watertight cable. Programming is done using a WiFi antenna. The electronics needs about one minute to boot after WiFi antenna is attached.

3.1.2 Connecting the Drive Unit

The connecting cable is connected to the drive unit by means of a plug connection. The connecting cable should not be disconnected.

The cable is provided with a 6-pin underwater socket on the control unit side and secured by a cap nut.

3.2 GENERAL MOUNTING

3.2.1 Changing the Battery

ATTENTION: The battery must be changed prior to each long-term mooring.

ATTENTION: To unscrew or tighten the screws, only use perfect, corrosion-free tools of bronze or special steel 1.4539 or titanium TiAl6V4.

To remove the pressure tube cover (piece with handle) of the control unit, the three hexagon screws M6 on the cover brim must be removed and screwed into the adjacent tap hole. Tighten the three hexagon screws **uniformly** by turning them. In this way, the cover is lifted out of its guide and can be removed completely. ***Please ensure that the cover is not canted during this procedure.***

Pull the battery out of the pressure tube and dispose it. Insert the new battery pack. Only battery packs manufactured by K.U.M. may be used. For long-term mooring, please observe the manufacturing date or the date of maturity.

Put the new **desiccant bags** (e.g. Minipax 1/6TME) into the electronic tube!

Slightly lubricate the O-rings in the grooves and the brim of the cover with silicone paste P.

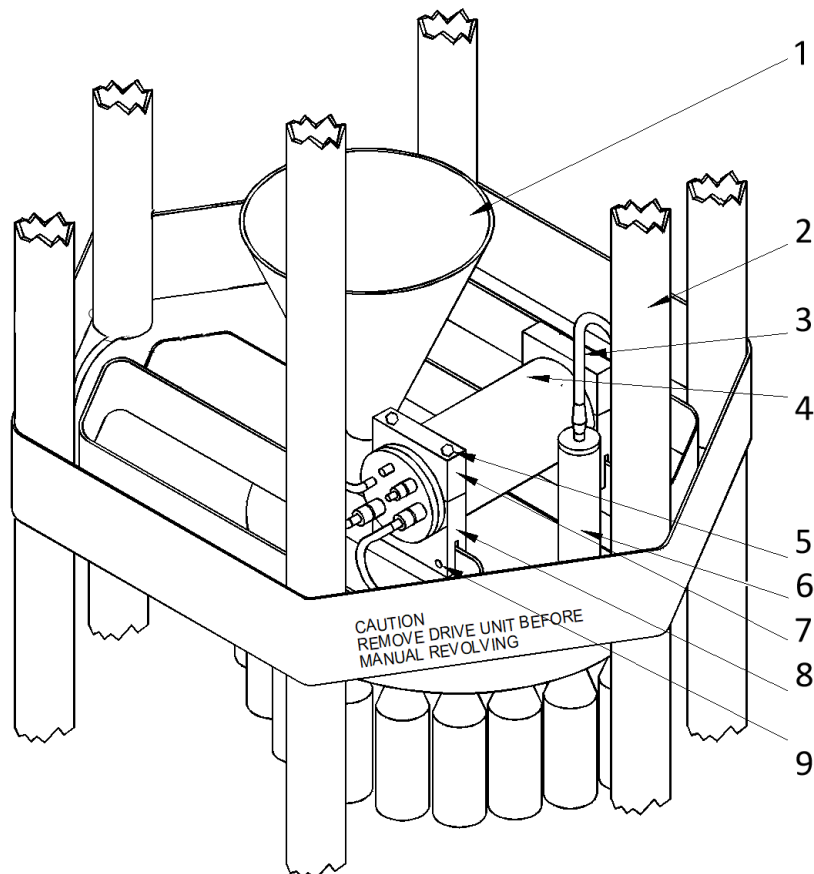
Put the cover of the control unit onto the pressure pipe, slightly lubricate the three hexagon screws with silicone paste P and fasten the cover with screws. Tighten the hexagon screws uniformly by turning them.

NOTE: Do not throw the empty batteries into the waste bin, but recycle them accordingly.

3.2.2 Mounting the Control Unit

Put the control unit into the bearing blocks (8) and fasten it with the clamp (7) and the hexagon screws (5).

ATTENTION: To unscrew or tighten the screws, only use perfect, corrosion-free tools of bronze or special steel 1.4539 or titanium TiAl6V4.



1 Funnel

2 Fastening spar for one-sided suspension

3 Connecting cable between control unit and drive unit

4 Control unit

5 Hexagon screws

6 Drive unit

7 Fastening clamp for control unit

8 Bearing blocks for control unit

9 Fastening holes for control unit

Fig. 12: Mounting the control unit

3.2.3 Mounting the Hand-Operated Revolving Mount

The hand-operated revolving mount has been designed to manually move the turntable easily and without problem. This is a simple revolving drive and is mounted instead of the drive unit. The drive unit has to be dismantled in this case.

The hand-operated revolving mount is inserted into the pinion and fastened by two screws. Now the turntable can be moved by a cam-bolt wrench or a ring-nut wrench of 13 mm width.

3.2.4 Mounting the Drive Unit

In order to be able to mount the drive unit, the ring gear has first to be positioned manually in such a manner that the intake opening of the sample bottles aligns with the funnel opening. "1" has to appear in the ring gear window (Fig. 14, part 3).

ATTENTION: To unscrew or tighten the screws, only use perfect, corrosion-free tools of bronze or special steel 1.4539 or titanium TiAl6V4.

Grease the cam slightly with silicone paste P and insert it into the drive opening with the ring gear facing down. Mount the drive unit onto the cam and fasten it with two screws (Fig. 14.3).

ATTENTION: When fastening, only use the included, corrosion-free screws.

3.2.5 Install the Sample Bottles

The sample bottles have a volume of 400 ml (approx. 14 oz.).

ATTENTION: In order to facilitate the deploying and recovering of the Sediment Trap, no sample bottle may be mounted under the funnel.

The first sampling position must be left vacant:

- when the funnel is open during deployment it fills with water more easy
- when the funnel is open during recovery the water inside the funnel can easily drain off.
There is no need to tilt the Trap for recovery

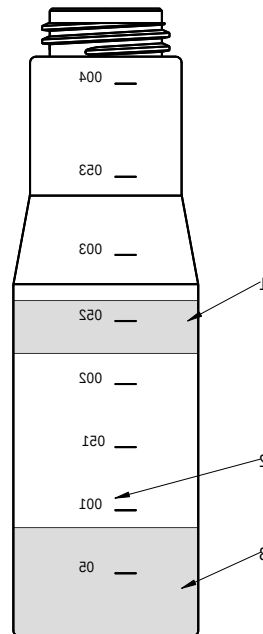
Install the sample bottles as follows:

Designate the sample bottles on the designation field (Fig. 13.1) and mark them unequivocally with a suitable water-proof felt-tip pen.

Unscrew the covers of the sample bottles, collect them and store them carefully.

ATTENTION: Fill the sample bottles with distilled water in order to avoid deformation due to water pressure!

One after the other, screw the sample bottles (not too tightly) into the turntable and check that they are fit securely. If the sample bottles are fit too tightly, leakage may result. The sampling positions are indicated by numbers on the ring gear. These are visible from inside through an opening (Fig. 14, 3) in the upper mounting plate below the funnel outlet.



1 Designation field

2 Scale for filling capacity

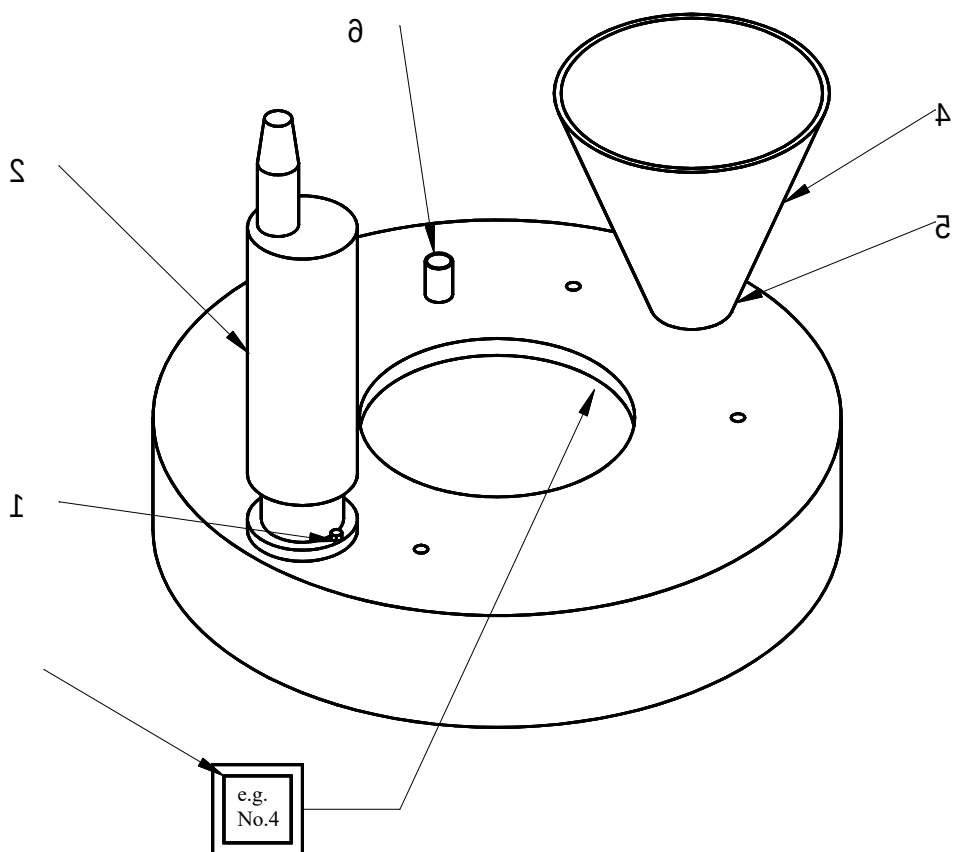
3 Handling piece

Fig. 13: Sample bottle

NOTE: Always ensure that the batteries are replaced by new ones after revolving with the motor unit and prior to deployment!

After removing the locking screw (Fig. 14.6) you will find an opening on top of the sample changer for filling the sample bottle with distilled water or fixative solution.

ATTENTION: Never turn the samples outside the water! Severe damage might be the result!



- 1 Screws for drive unit
- 2 Drive unit
- 3 Sample No. indicator

- 4 Funnel
- 5 Filler tube
- 6 Locking screw

Fig. 14: View of sample changer

3.2.6 Remove the Sample Bottles

Manually unscrew the sample bottles from the turntable and remove them carefully downward.

Close the sample bottles with the screw caps provided for this purpose.

Further measures are not required. The sample changer is ready for the next measuring cycle **after cleaning, programming and changing the battery.**

Additional sample bottles can be obtained as accessories (see section 5.3.2).

4. OPERATION AND PROGRAMMING

4.1 GENERAL

We recommend that you keep a deploying protocol for each Sediment Trap along with a logbook containing the data of the deploying, the recovery and maintenance work.

This log must also document the inspection, which should be performed by the manufacturer every five years.

The internal webpage allows you to print out configuration, mooring and recovery protocols.

4.2 IMPORTANT INSTRUCTIONS FOR HANDLING THE WEBPAGE

The webpage enables you to create a logfile prior to deploying and a further logfile after recovery that can be referred to for trouble-shooting in case of malfunctions.

Caution: Prior to shipping the Sediment Trap to the manufacturer because of a malfunction, all of the files generated during the programming and after recovery must be sent to the manufacturer for examination. Only then the cause of the trouble can be sufficiently judged, especially in case of warranty demands.

4.3 GENERAL OPERATING SEQUENCE

4.3.1 Procedure to Deploy the Sediment Trap

This section includes a short instruction (check list) of the general procedure for deploying the Sediment Trap. The programming dialogue guides you through the programming process. The following processes must be performed in the sequence given below before deploying the Sediment Trap:

Process	Section
1. Install the sample bottles.....	3.2.5
2. Changing the Battery.....	3.2.1
3. Mounting and connecting the control unit and the drive unit	3.2.2, 3.2.4
4. Connecting the WiFi antenna to the control unit	
5. Enter the webpage of the unit, IP address 10.0.0.1	see attachment
6. Entering the name of the Sediment Trap	see attachment
7. Setting up the configuration and programming of the sampling data.....	see attachment
8. Transmitting the sampling data to the electronics	see attachment
9. Disconnecting the antenna and applying dummy connector	
10. Deploying the Sediment Trap	

4.3.2 Procedure to Recover the Sediment Trap

This section includes a short instruction (check list) of the general procedure to recover the Sediment Trap. To recover the Sediment Trap the following processes must be performed in the sequence given below:

Process	Section
1. Recover the Sediment Trap and rough cleaning by rinsing with fresh water	
2. Connect WiFi antenna to the control unit	
3. Enter the webpage of the unit, IP address 10.0.0.1	see attachment
4. Save the data stored in the Sediment Trap.....	see attachment
5. Disconnect the antenna and apply dummy connector	
6. Remove the sample bottles.....	3.2.6
7. Final care and maintenance	5.2

5. MAINTENANCE AND CARE

5.2 MAINTENANCE AND CARE

After each deployment the Sediment Trap *must* be rinsed thoroughly with fresh water! If necessary, it must be completely freed from mussels and fouling. The plug connectors of the drive unit, as well as the pinion of the sample changer must be cleaned from seawater remainders and slightly greased with silicone paste P.

Care of the components is restricted to the cleaning of the casing surfaces and check for outer damage. When cleaning, it is advisable to remove the dirt with a brush and a wetted cloth. Generally, the use of soft soap suds will be sufficient.

ATTENTION: Do not use strong cleaning agents or solvents (gasoline).

5.2.1 Putting out of Operation

In order to put the Sediment Trap out of operation for a longer period, the following measures must be performed:

- Remove the batteries from the battery holders (see section 3.2.1) to avoid leakage.
- Protect the plug connectors on the control unit and the drive unit against dust and humidity. If the units are stored separately, provide them with protective caps or connect the cables correctly.

When putting it into operation again, new batteries must be inserted (see section 3.2.1).

5.2.2 Transport and Storage

The Sediment Trap must be transported and stored appropriately. It must be protected against detrimental influences and heavy vibrations such as shock, impact, etc.

During storage the batteries must be removed (danger of leakage) (see section 3.2.1).

The storage temperature should remain within -30 °C to +70 °C.

5.2.4 Disassembling and Cleaning the Sample Changer

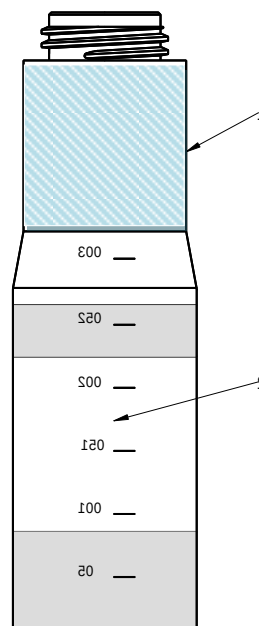
In order to maintain the sample changer in case of contamination or in case of failure, it can be disassembled and reassembled after cleaning as follows.

Dismount the control unit and the drive unit.

In order to dismount the sample changer, unscrew the four hexagon screws from the central intermediate bottom using a 13 mm wrench. In order to do this, hold the sample changer from below with the aid of a second person or support it if necessary. The sample changer has a weight of approx. 14 kg.

Put the sample changer onto a strong base plate with the openings of the sample bottles showing upwards.

Slip the mounting aid (Fig. 28.1) onto all sample bottles and screw all sample bottles into the sample changer by hand. The pressure springs are fixed and thus relieve the turntable.



- 1 Mounting aid
- 2 Sample bottle

Figure 28: Sample bottle with mounting aid

Remove the internal ring gear (Fig 3.9) together with the six slide pieces from the sample changer. In order to do this, the two hexagon screws fixing the internal ring gear to the sample changer are unscrewed with a 13 mm wrench.

Dismount the six slide pieces (Fig. 3.6) by removing the screws. The ring gear (Fig. 3.4) can now be removed from the mounting plate.

Turn over the mounting plate and remove the three screws that are screwed in from the top to fasten the slide ring. Remove the slide ring.

In order to be able to disassemble the ring gear, the sample bottles must be unscrewed. **You must proceed carefully, because the pressure spring releases the tension.** Hold the bottle holder by hand in order to avoid deformation.

All components should only be cleaned using soft soap suds and can be disinfected with a disinfectant when necessary.

ATTENTION: Do not use strong cleaning agents or solvents (gasoline). Do not treat or scratch sliding surfaces and ring gear by sharp objects, e.g. a screw driver.

The reassembly is performed in reverse order.

5.3 SCOPE OF DELIVERY AND ACCESSORIES

5.3.1. Scope of Delivery

The scope of delivery of the Sediment Trap includes:

- Frame construction with intermediate bottom
- Funnel
- Sample changer with 20 sample bottles
- Control unit, optional with Inclinator
- Drive unit with cable
- User's manual "SEDIMENT TRAP"

5.3.2. List of Accessories

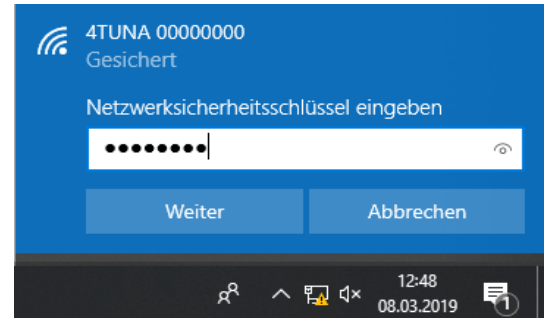
The following accessories can be obtained via the type number from K.U.M. Umwelt- und Meerestechnik GmbH:

Designation	Typ
Maintenance set, including - O-rings for the drive shaft - O-rings for the battery cover - Silicone paste P (small quantity) - Dessiccant bag	K/MT 310-01
Battery set, 12 pieces alkaline mono type	K/MT 310-02
Set of tools needed for maintenance	K/MT 310-03
Electronics of the control unit, complete with cover	K/MT 310-04
Motor-gear unit, complete with rotation sensor	K/MT 310-05
Mounting set for sample changer with mounting aids for sample bottles	K/MT 310-06
Repair set, including - Electronics of control unit, complete with cover - Motor-gear unit - Service manual - Maintenance set (3x) - Set of screws of titanium - Tools set	K/MT 236-07
Protective cap	K/MT 320-02
Hand-operated revolving mount	K/MT 320-03
Crowfoot fastening, complete	K/MT 320-04
Service manual	K/MT 320-05
Sample bottle, 1 piece	K/MT 320-06

6. PROGRAMMING THE SEDIMENT TRAP

6.1. ESTABLISH WIFI-CONNECTION

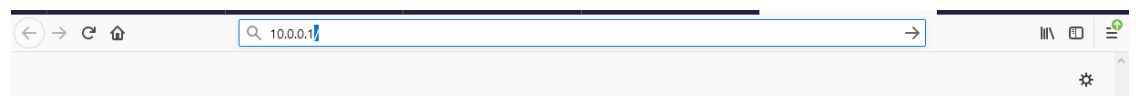
Remove the dummy connector from the electronics pressure tube and attach the WiFi connector. Make sure the electronics is powered. After one minute a WiFi network will appear. The network is called '4TUNA xxxxx' where xxxxx is replaced by the serial number of the electronics.



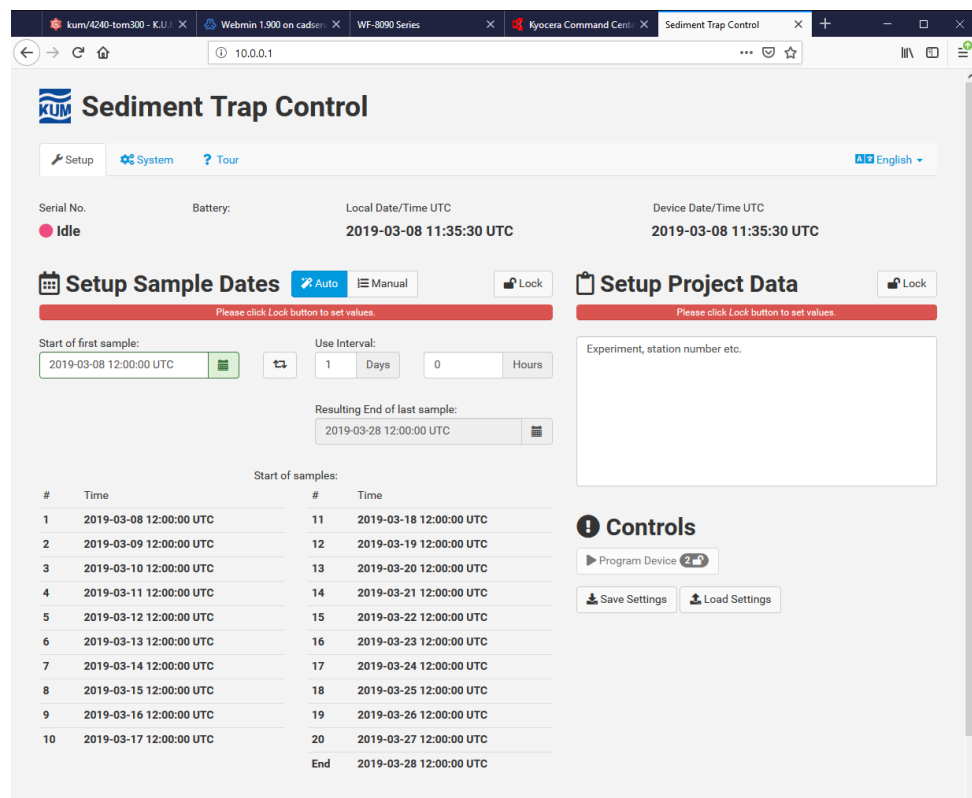
Enter the password and connect to the web-server. The default password is 'seismics' and it can be changed after accessing the website.

6.2. OPEN WEBPAGE

Open a browser and enter '10.0.0.1' in the address line:

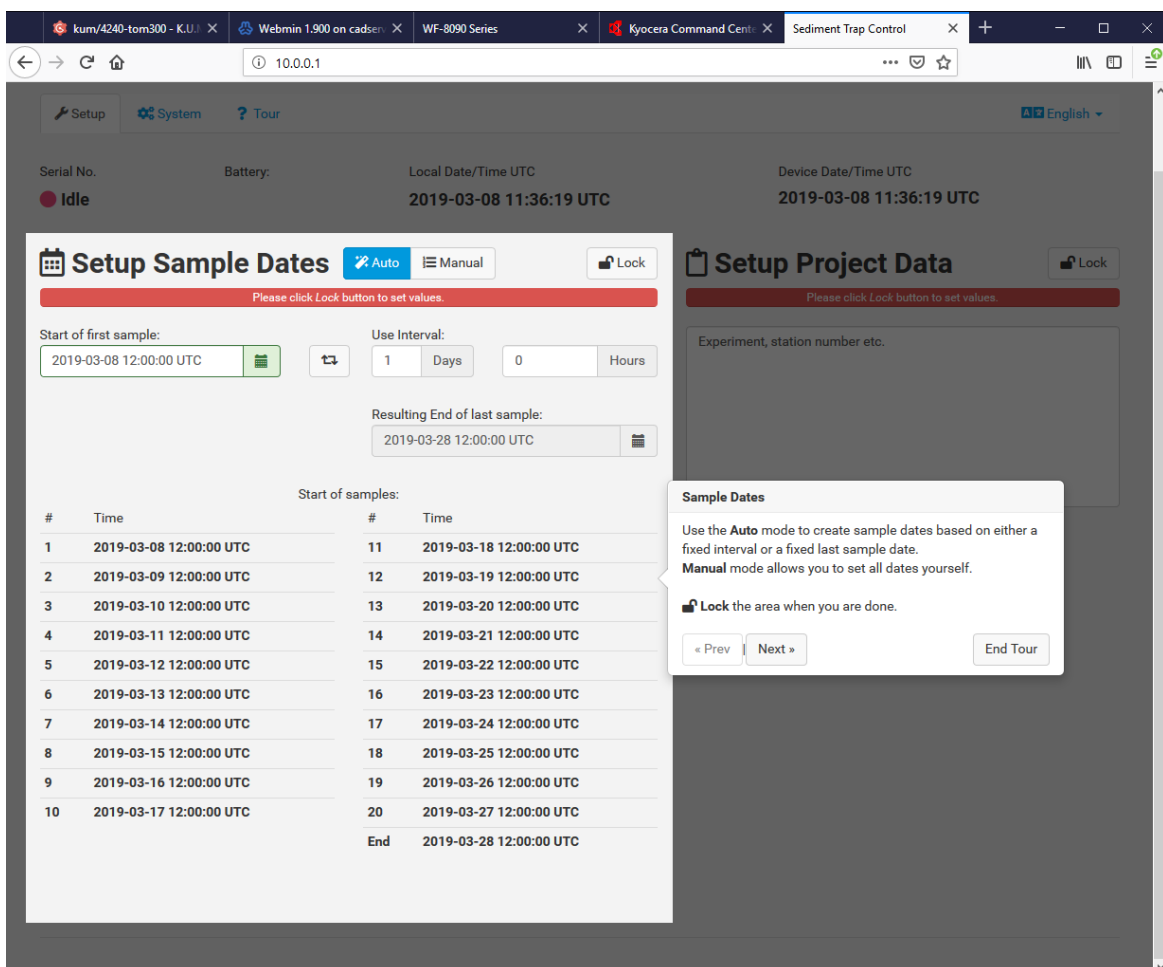
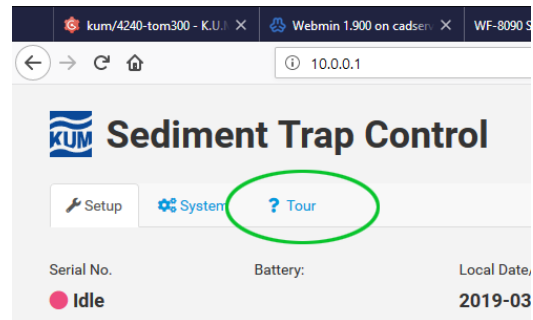


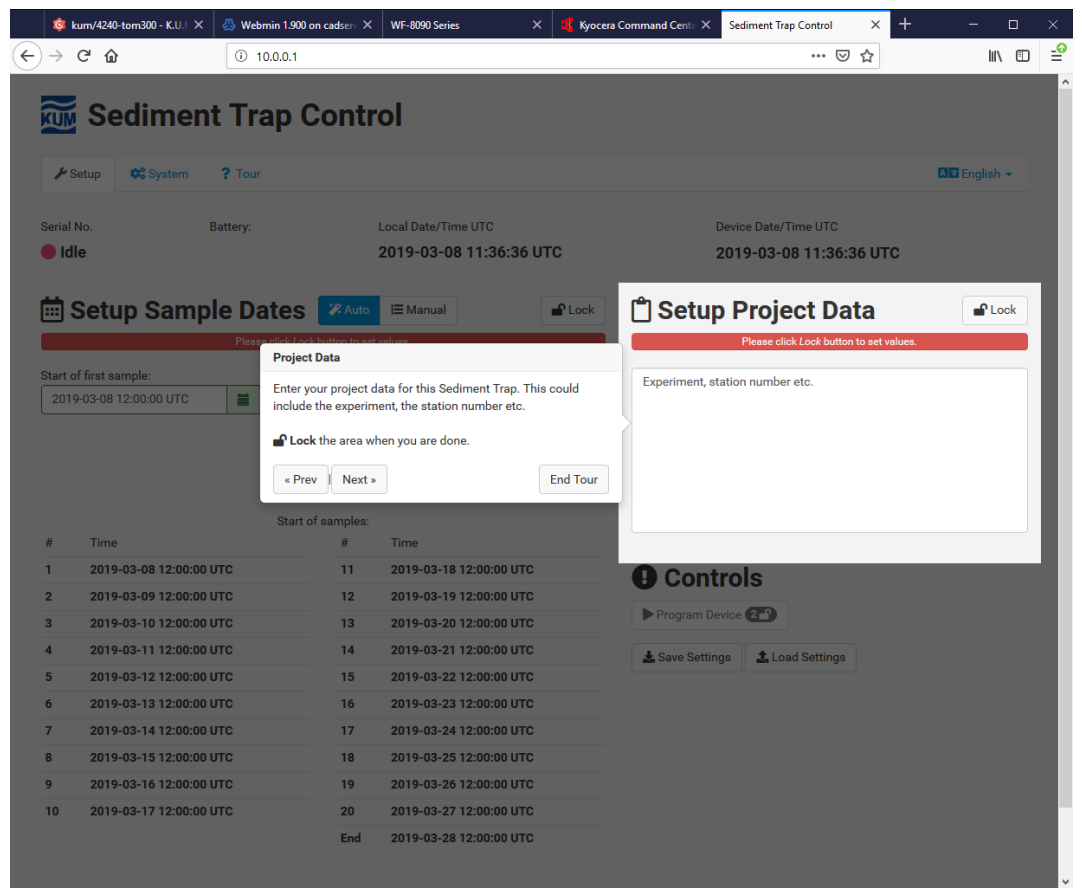
The web-page opens:



6.3. PROGRAMMING THE SEDIMENT TRAP

Easiest way to get familiar with programming the Trap is watching the tour. Click on 'Tou' as shown in the picture right, then a tour as below will start:





Sediment Trap Control

Serial No. **Idle** Battery: Local Date/Time UTC **2019-03-08 11:36:36 UTC** Device Date/Time UTC **2019-03-08 11:36:36 UTC**

Setup Sample Dates **Auto** **Manual** **Lock**

Please click Lock button to set values.

Start of first sample: **2019-03-08 12:00:00 UTC**

Project Data

Enter your project data for this Sediment Trap. This could include the experiment, the station number etc.

Lock the area when you are done.

« Prev Next » End Tour

Setup Project Data **Lock**

Please click Lock button to set values.

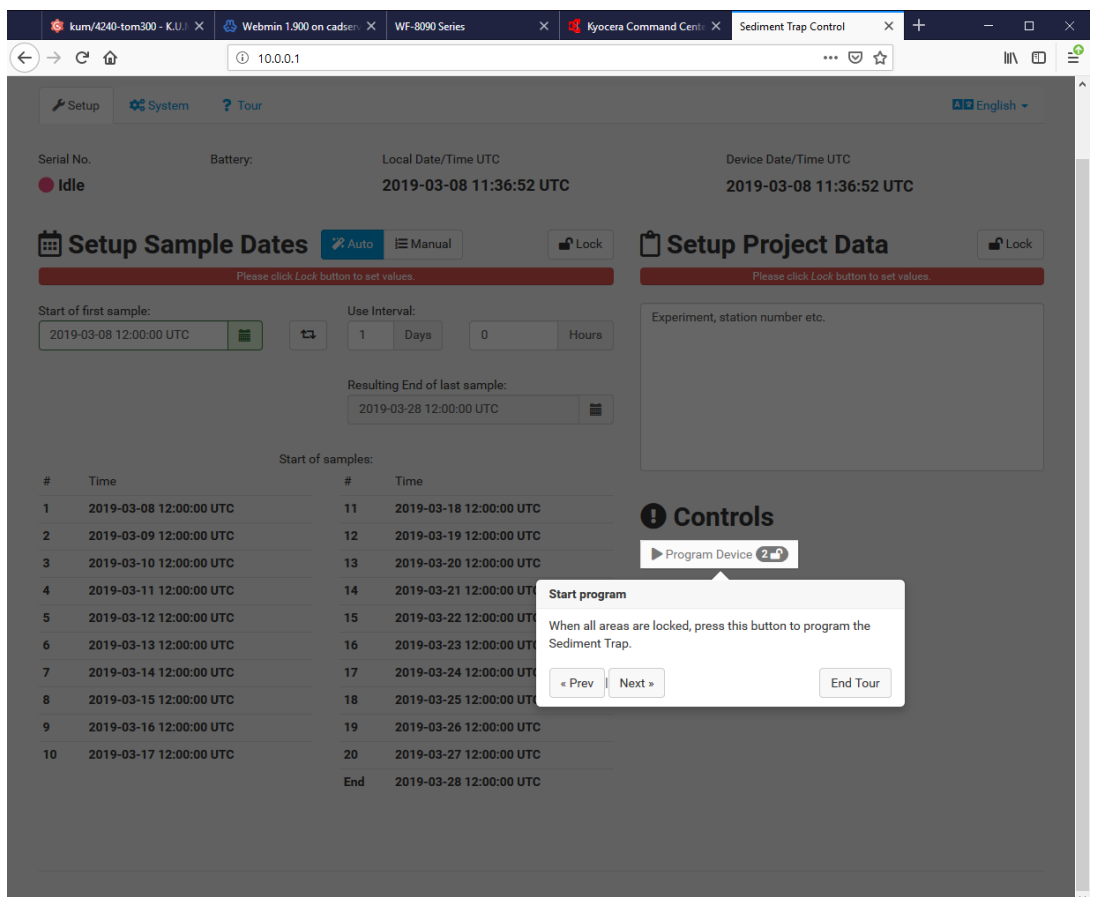
Experiment, station number etc.

Controls

Program Device **2.0**

Save Settings Load Settings

#	Time	#	Time
1	2019-03-08 12:00:00 UTC	11	2019-03-18 12:00:00 UTC
2	2019-03-09 12:00:00 UTC	12	2019-03-19 12:00:00 UTC
3	2019-03-10 12:00:00 UTC	13	2019-03-20 12:00:00 UTC
4	2019-03-11 12:00:00 UTC	14	2019-03-21 12:00:00 UTC
5	2019-03-12 12:00:00 UTC	15	2019-03-22 12:00:00 UTC
6	2019-03-13 12:00:00 UTC	16	2019-03-23 12:00:00 UTC
7	2019-03-14 12:00:00 UTC	17	2019-03-24 12:00:00 UTC
8	2019-03-15 12:00:00 UTC	18	2019-03-25 12:00:00 UTC
9	2019-03-16 12:00:00 UTC	19	2019-03-26 12:00:00 UTC
10	2019-03-17 12:00:00 UTC	20	2019-03-27 12:00:00 UTC
		End	2019-03-28 12:00:00 UTC



Sediment Trap Control

Serial No. **Idle** Battery: Local Date/Time UTC **2019-03-08 11:36:52 UTC** Device Date/Time UTC **2019-03-08 11:36:52 UTC**

Setup Sample Dates **Auto** **Manual** **Lock**

Please click Lock button to set values.

Start of first sample: **2019-03-08 12:00:00 UTC**

Use Interval: **1** Days **0** Hours

Resulting End of last sample: **2019-03-28 12:00:00 UTC**

Setup Project Data **Lock**

Please click Lock button to set values.

Experiment, station number etc.

Controls

Program Device **2.0**

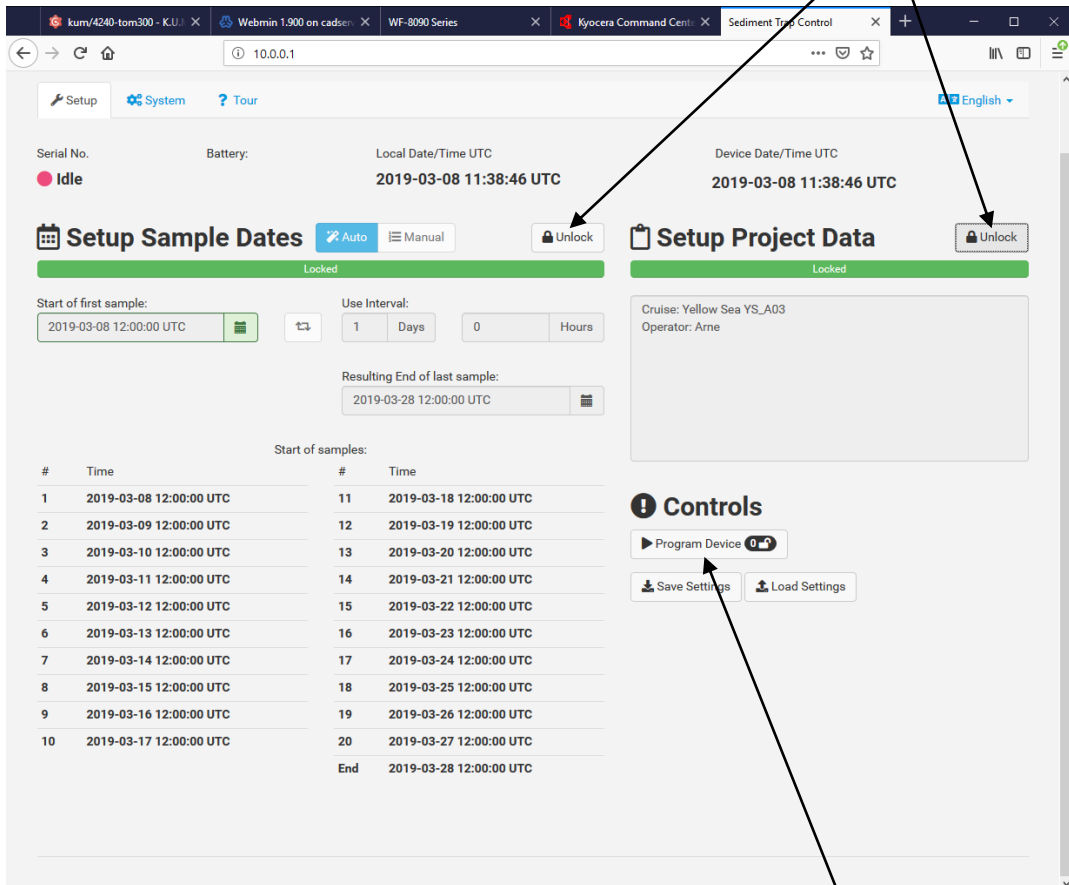
Start program

When all areas are locked, press this button to program the Sediment Trap.

« Prev Next » End Tour

#	Time	#	Time
1	2019-03-08 12:00:00 UTC	11	2019-03-18 12:00:00 UTC
2	2019-03-09 12:00:00 UTC	12	2019-03-19 12:00:00 UTC
3	2019-03-10 12:00:00 UTC	13	2019-03-20 12:00:00 UTC
4	2019-03-11 12:00:00 UTC	14	2019-03-21 12:00:00 UTC
5	2019-03-12 12:00:00 UTC	15	2019-03-22 12:00:00 UTC
6	2019-03-13 12:00:00 UTC	16	2019-03-23 12:00:00 UTC
7	2019-03-14 12:00:00 UTC	17	2019-03-24 12:00:00 UTC
8	2019-03-15 12:00:00 UTC	18	2019-03-25 12:00:00 UTC
9	2019-03-16 12:00:00 UTC	19	2019-03-26 12:00:00 UTC
10	2019-03-17 12:00:00 UTC	20	2019-03-27 12:00:00 UTC
		End	2019-03-28 12:00:00 UTC

Follow the instructions and set up the sample times. In case you enter invalid times (e.g. delay between two samples too short, date that already passed) the software will mark it red and warn. Only if all entries are valid you will be able to lock your data by clicking the lock symbol on top of the red bar which will turn into green if succeeded.



Serial No. **Idle** Battery: Local Date/Time UTC **2019-03-08 11:38:46 UTC** Device Date/Time UTC **2019-03-08 11:38:46 UTC**

Setup Sample Dates Auto Manual Unlock **Setup Project Data** Unlock

Start of first sample: Use Interval: Days Hours

Resulting End of last sample:

#	Time	#	Time
1	2019-03-08 12:00:00 UTC	11	2019-03-18 12:00:00 UTC
2	2019-03-09 12:00:00 UTC	12	2019-03-19 12:00:00 UTC
3	2019-03-10 12:00:00 UTC	13	2019-03-20 12:00:00 UTC
4	2019-03-11 12:00:00 UTC	14	2019-03-21 12:00:00 UTC
5	2019-03-12 12:00:00 UTC	15	2019-03-22 12:00:00 UTC
6	2019-03-13 12:00:00 UTC	16	2019-03-23 12:00:00 UTC
7	2019-03-14 12:00:00 UTC	17	2019-03-24 12:00:00 UTC
8	2019-03-15 12:00:00 UTC	18	2019-03-25 12:00:00 UTC
9	2019-03-16 12:00:00 UTC	19	2019-03-26 12:00:00 UTC
10	2019-03-17 12:00:00 UTC	20	2019-03-27 12:00:00 UTC
		End	2019-03-28 12:00:00 UTC

Controls

Program Device Save Settings Load Settings

Next, enter your Project data and lock it by clicking of the lock symbols.

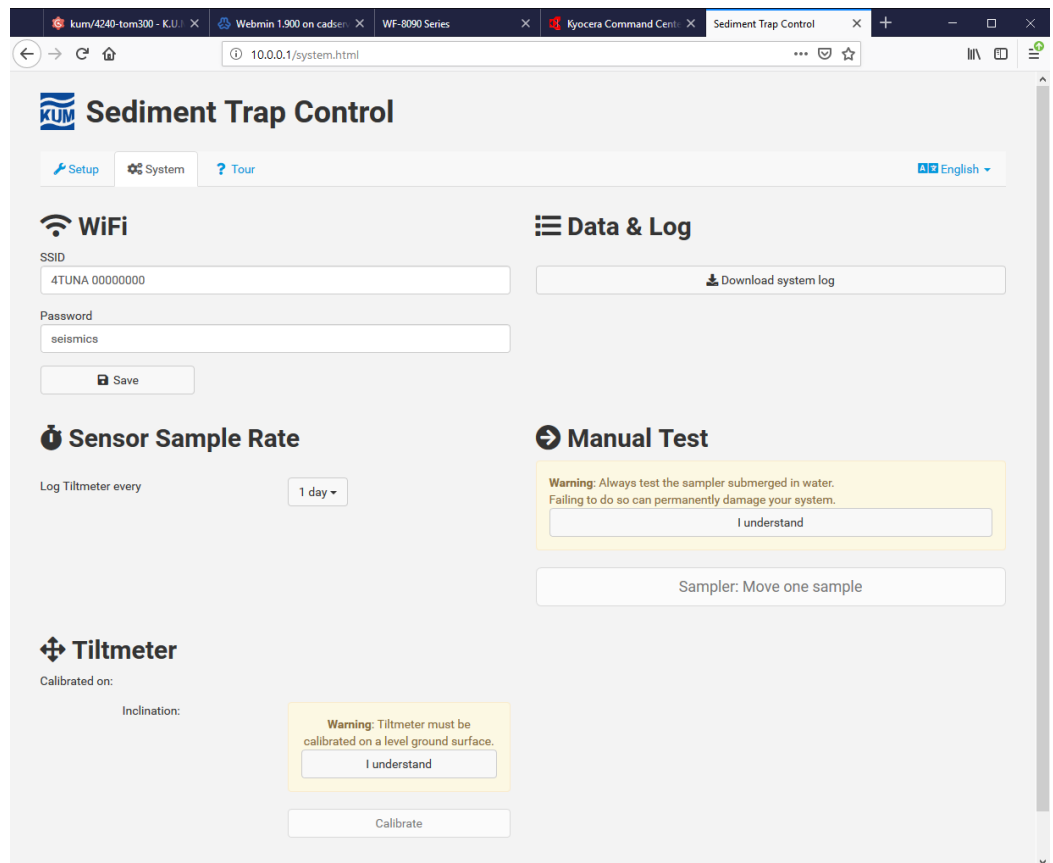
The red bars turn green.

You are now ready to send your data to the Sediment Trap by clicking 'Program Device'.

6.4. 'SYSTEM': TILTMETER AND MANUAL TEST

On the second tab 'System' you can calibrate your tiltmeter and define the tiltmeter sampling rate.

Attention: For a 12 month deployment the sampling rate of the tiltmeter must be set to '1 day'.

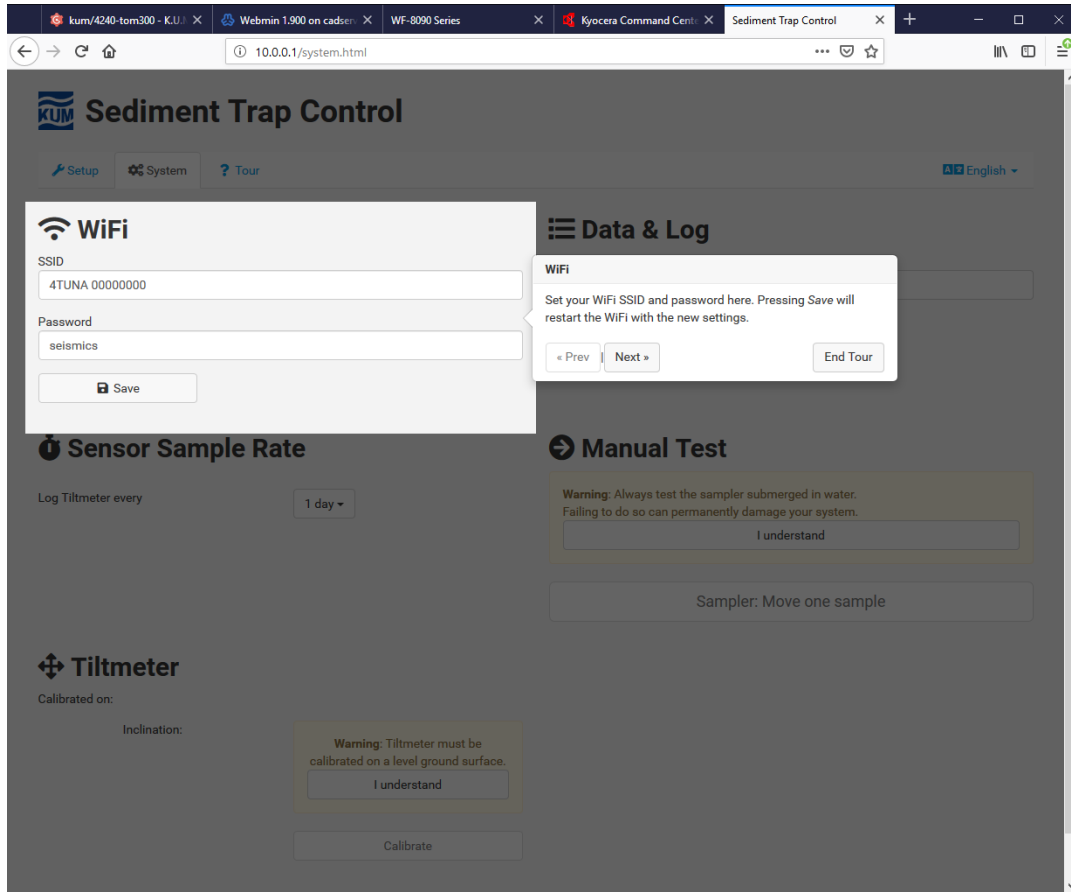


The screenshot shows the 'Sediment Trap Control' web interface. The 'System' tab is active, displaying settings for WiFi, Sensor Sample Rate, and Tiltmeter. The WiFi section shows SSID '4TUNA 00000000' and Password 'seismics'. The Sensor Sample Rate section shows 'Log Tiltmeter every' set to '1 day'. The Tiltmeter section shows 'Calibrated on:' and 'Inclination:' fields. A 'Manual Test' section contains a warning: 'Warning: Always test the sampler submerged in water. Failing to do so can permanently damage your system.' and a button 'I understand'. A 'Calibrate' button is also present.

You can also force a manual turn of the turntable using 'Manual Test'.

ATTENTION: Never turn the samples outside the water! Severe damage might be the result!

This tab's tour is shown on the following pages.



Sediment Trap Control

Setup System Tour

English

WiFi

SSID: 4TUNA 00000000

Password: seismics

Save

Data & Log

WiFi

Set your WiFi SSID and password here. Pressing Save will restart the WiFi with the new settings.

« Prev Next » End Tour

Sensor Sample Rate

Log Tiltmeter every 1 day

Tiltmeter

Calibrated on:

Inclination:

Warning: Tiltmeter must be calibrated on a level ground surface.

I understand

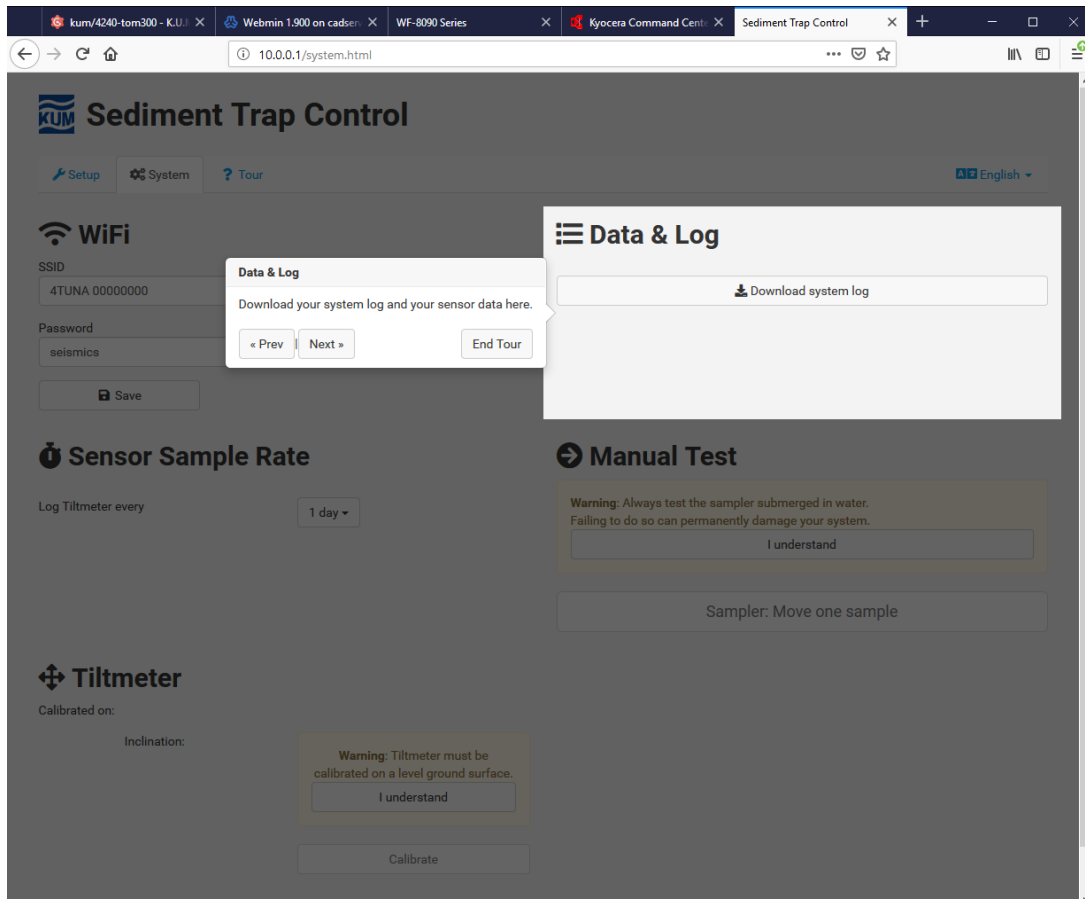
Calibrate

Manual Test

Warning: Always test the sampler submerged in water. Failing to do so can permanently damage your system.

I understand

Sampler: Move one sample



Sediment Trap Control

Setup System Tour

English

WiFi

SSID: 4TUNA 00000000

Password: seismics

Save

Data & Log

Data & Log

Download your system log and your sensor data here.

« Prev Next » End Tour

Download system log

Sensor Sample Rate

Log Tiltmeter every 1 day

Tiltmeter

Calibrated on:

Inclination:

Warning: Tiltmeter must be calibrated on a level ground surface.

I understand

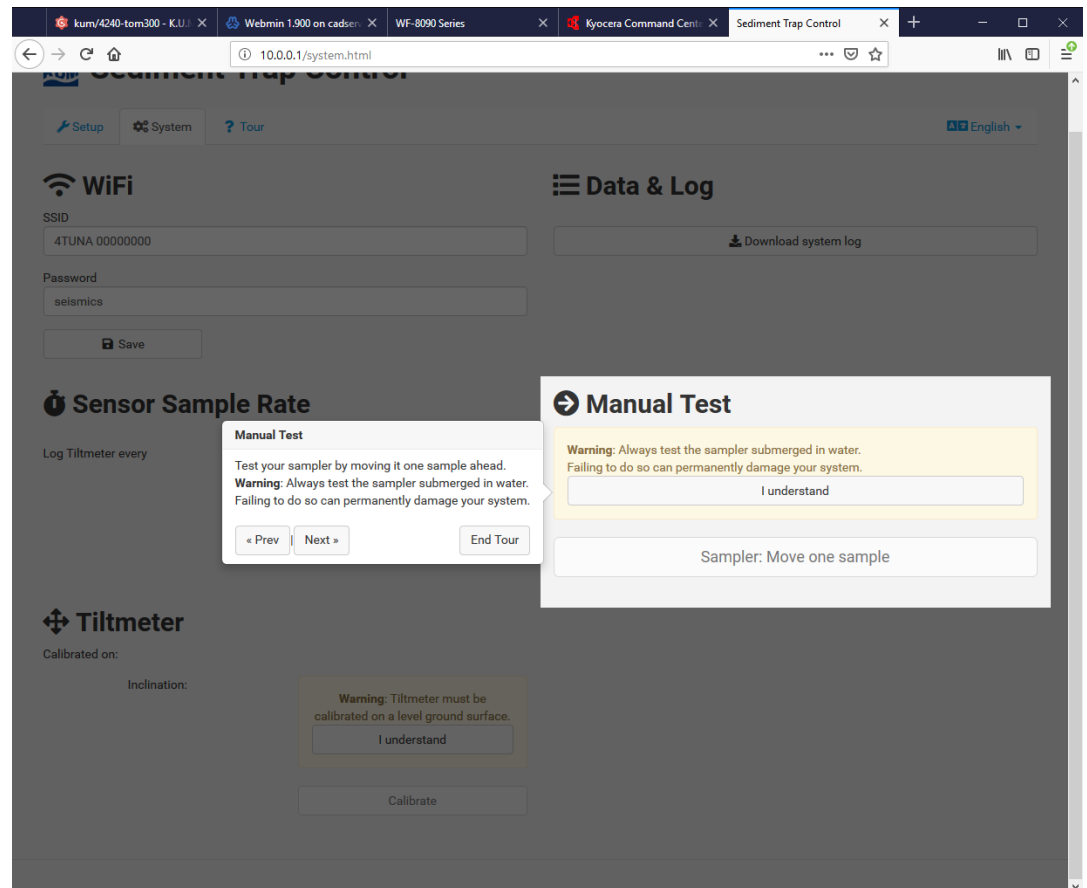
Calibrate

Manual Test

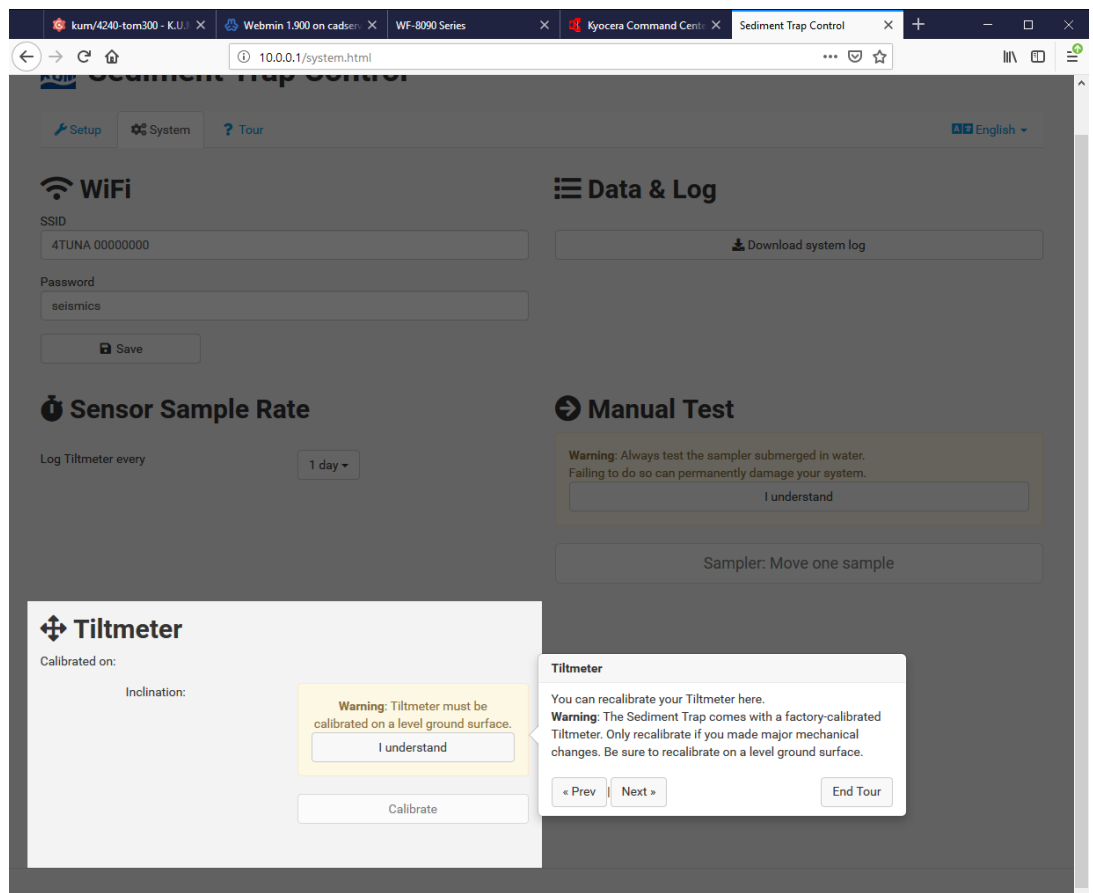
Warning: Always test the sampler submerged in water. Failing to do so can permanently damage your system.

I understand

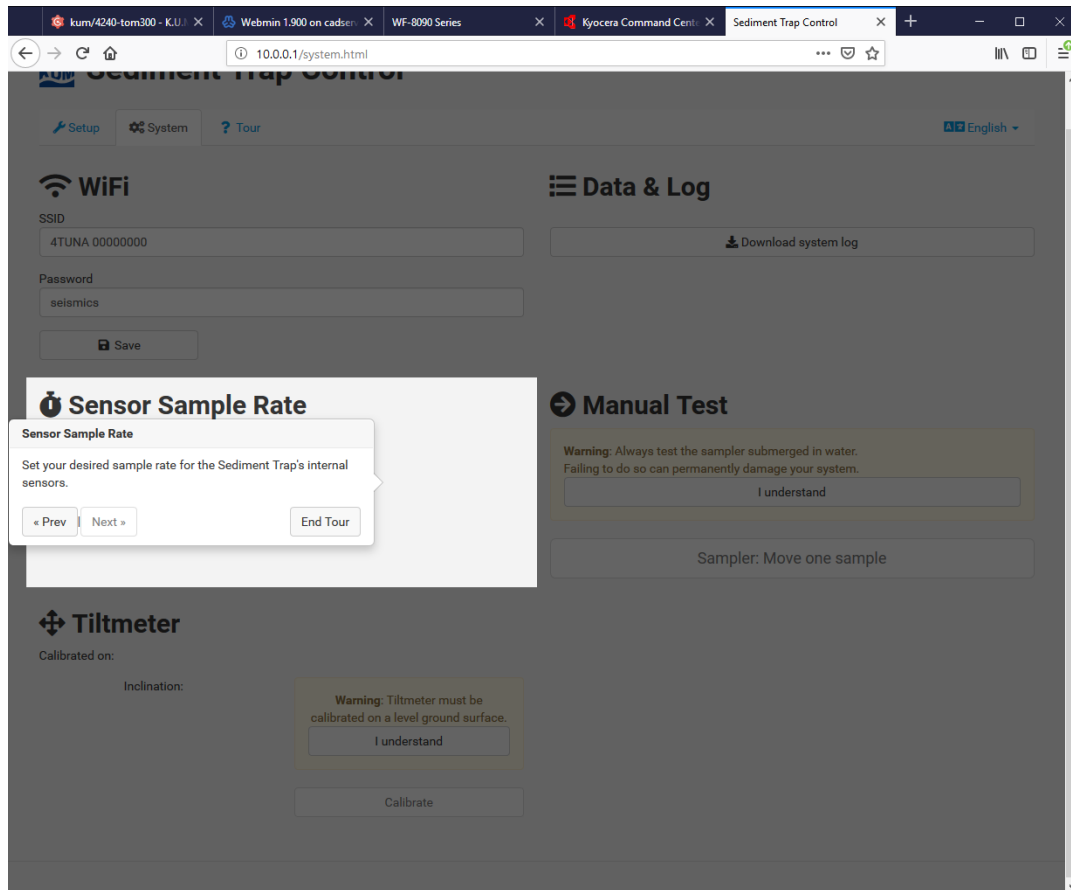
Sampler: Move one sample



The screenshot shows the 'Sediment Trap Control' web interface. The 'WiFi' section has fields for SSID (4TUNA 00000000) and Password (seismics), with a 'Save' button. The 'Sensor Sample Rate' section has a 'Log Tiltmeter every' dropdown set to '1 day'. A 'Manual Test' dialog box is open, showing a warning: 'Warning: Always test the sampler submerged in water. Failing to do so can permanently damage your system.' Below the warning is an 'I understand' button. The 'Tiltmeter' section has a 'Calibrated on:' field and a 'Calibrate' button. A 'Data & Log' section has a 'Download system log' button.



The screenshot shows the 'Sediment Trap Control' web interface. The 'Tiltmeter' section has a 'Calibrated on:' field and a 'Calibrate' button. A 'Tiltmeter' dialog box is open, showing a warning: 'Warning: Tiltmeter must be calibrated on a level ground surface. Failing to do so can permanently damage your system.' Below the warning is an 'I understand' button. The 'Manual Test' section has a 'Warning: Always test the sampler submerged in water. Failing to do so can permanently damage your system.' and an 'I understand' button. The 'Sensor Sample Rate' section has a 'Log Tiltmeter every' dropdown set to '1 day'. The 'Data & Log' section has a 'Download system log' button.



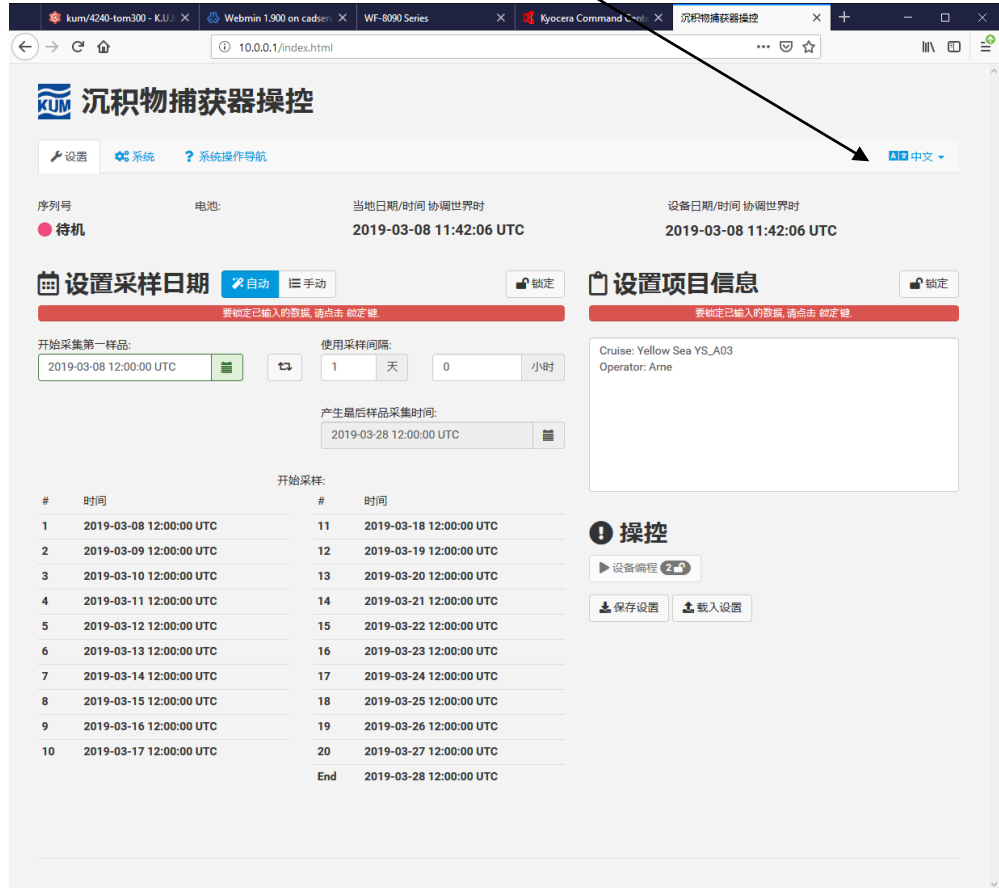
The screenshot shows a web browser window with multiple tabs. The active tab is titled "Sediment Trap Control" and displays the URL "10.0.0.1/system.html". The interface is dark-themed and features a top navigation bar with links for "Setup", "System", and "Tour". A language selector for "English" is also present.

The main content area is divided into several sections:

- WiFi:** Includes fields for "SSID" (containing "4TUNA 00000000") and "Password" (containing "seismics"), with a "Save" button below.
- Data & Log:** Features a "Download system log" button.
- Sensor Sample Rate:** A modal dialog box is open, displaying the title "Sensor Sample Rate" and the instruction "Set your desired sample rate for the Sediment Trap's internal sensors." It includes "Prev", "Next", and "End Tour" buttons.
- Manual Test:** Contains a warning message: "Warning: Always test the sampler submerged in water. Failing to do so can permanently damage your system." Below the warning is an "I understand" button. At the bottom of this section is a button labeled "Sampler: Move one sample".
- Tiltmeter:** Includes a "Calibrated on:" label, an "Inclination:" label, a warning message: "Warning: Tiltmeter must be calibrated on a level ground surface." with an "I understand" button, and a "Calibrate" button.

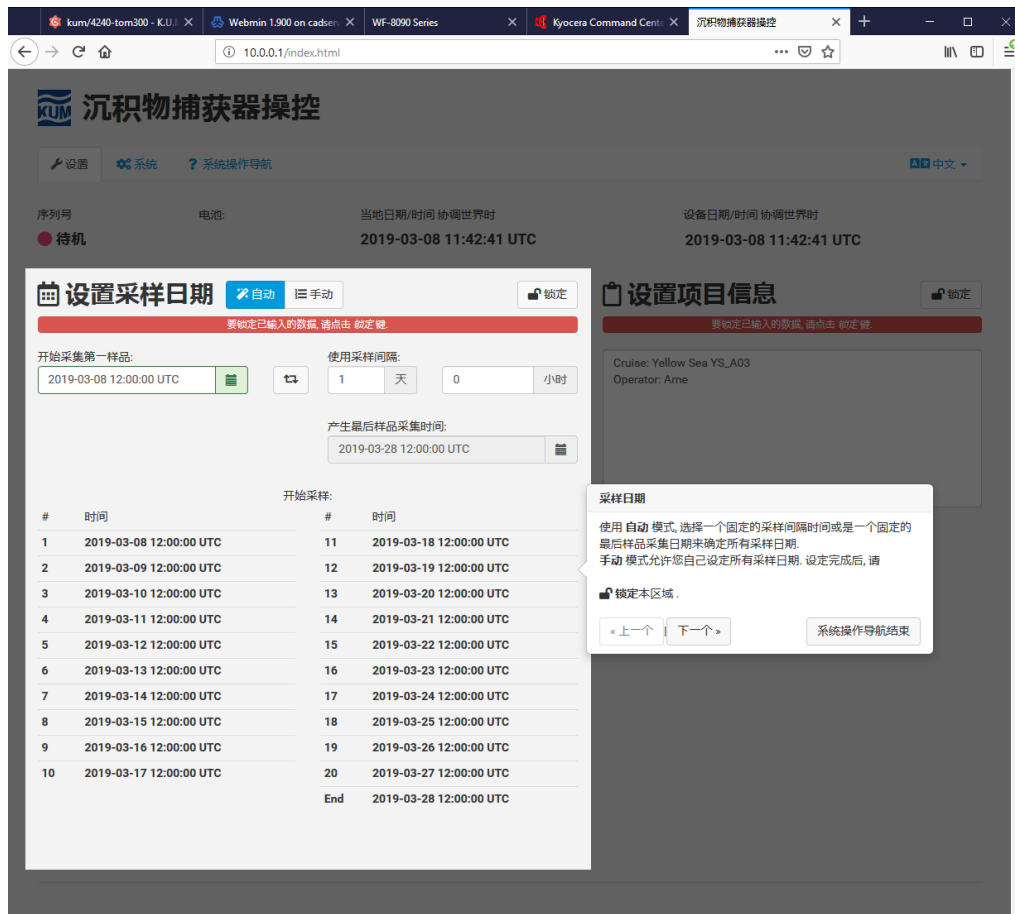
6.5. LANGUAGE PACK

Language can be set by clicking the language symbol. Both webpages and tours will change to Chinese.



The screenshot displays the '沉积物捕获器操控' (Sediment Collector Control) web interface. The top navigation bar includes '设置' (Settings), '系统' (System), and '系统操作导航' (System Operation Navigation). The main content area is divided into several sections:

- 系统状态 (System Status):** Shows the device status as '待机' (Standby), battery level, and local/world time (2019-03-08 11:42:06 UTC).
- 设置采样日期 (Set Sampling Date):** Includes buttons for '自动' (Automatic) and '手动' (Manual), and a '锁定' (Lock) button. It displays the start time (2019-03-08 12:00:00 UTC) and the interval (1 day, 0 hours).
- 设置项目信息 (Set Project Information):** Includes a '锁定' (Lock) button and displays 'Cruise: Yellow Sea YS_A03' and 'Operator: Ame'.
- 开始采样 (Start Sampling):** A table listing sampling times from 2019-03-08 12:00:00 UTC to 2019-03-28 12:00:00 UTC.
- 操控 (Control):** Includes buttons for '设备编程' (Device Programming), '保存设置' (Save Settings), and '载入设置' (Load Settings).



沉积物捕获器操控

序列号: 待机 电池: 2019-03-08 11:42:41 UTC 设备日期/时间 协调世界时: 2019-03-08 11:42:41 UTC

设置采样日期 (自动 | 手动) [锁定]

要锁定已输入的数据, 请点击 锁定键。

开始采集第一样品: 2019-03-08 12:00:00 UTC [日历图标]

使用采样间隔: 1 天 0 小时

产生最后样品采集时间: 2019-03-28 12:00:00 UTC [日历图标]

开始采样:

#	时间	#	时间
1	2019-03-08 12:00:00 UTC	11	2019-03-18 12:00:00 UTC
2	2019-03-09 12:00:00 UTC	12	2019-03-19 12:00:00 UTC
3	2019-03-10 12:00:00 UTC	13	2019-03-20 12:00:00 UTC
4	2019-03-11 12:00:00 UTC	14	2019-03-21 12:00:00 UTC
5	2019-03-12 12:00:00 UTC	15	2019-03-22 12:00:00 UTC
6	2019-03-13 12:00:00 UTC	16	2019-03-23 12:00:00 UTC
7	2019-03-14 12:00:00 UTC	17	2019-03-24 12:00:00 UTC
8	2019-03-15 12:00:00 UTC	18	2019-03-25 12:00:00 UTC
9	2019-03-16 12:00:00 UTC	19	2019-03-26 12:00:00 UTC
10	2019-03-17 12:00:00 UTC	20	2019-03-27 12:00:00 UTC
		End	2019-03-28 12:00:00 UTC

设置项目信息 [锁定]

要锁定已输入的数据, 请点击 锁定键。

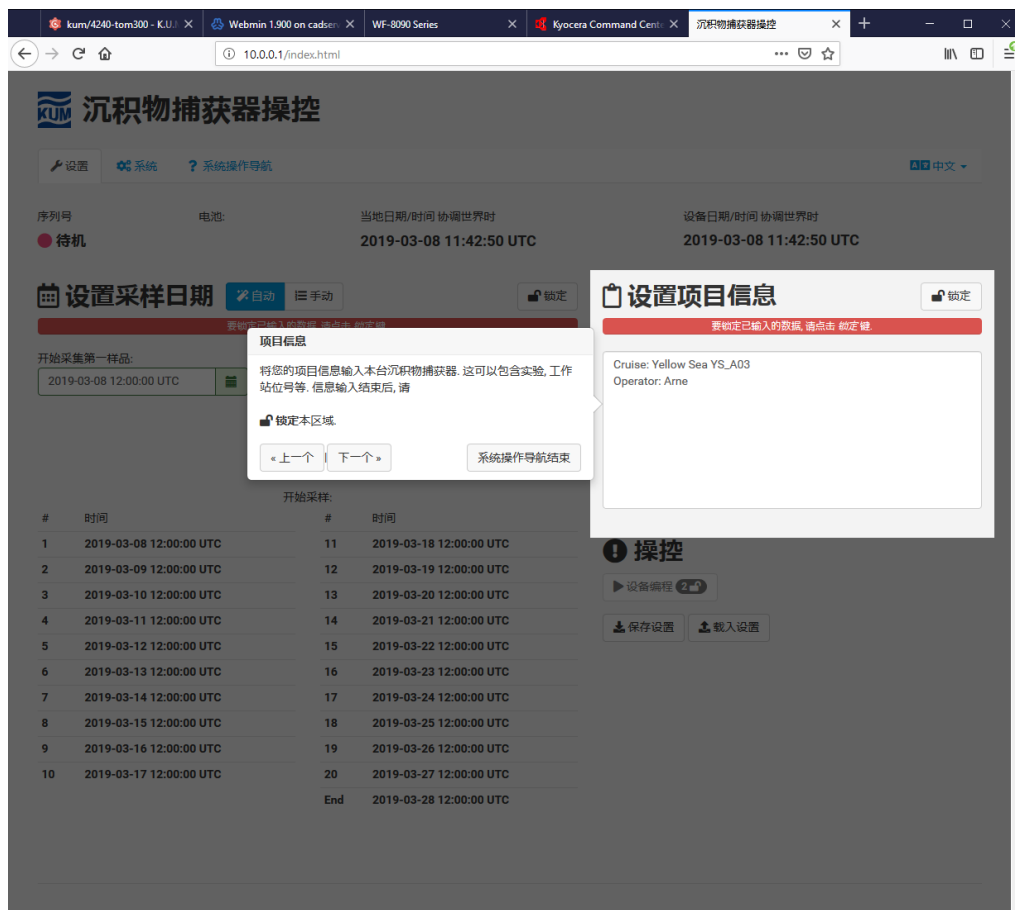
Cruise: Yellow Sea YS_A03
Operator: Arne

采样日期

使用 自动 模式, 选择一个固定的采样间隔时间或是一个固定的最后样品采集日期来确定所有采样日期。
手动 模式允许您自己设定所有采样日期, 设定完成后, 请

[锁定本区域]

« 上一个 | 下一个 » [系统操作导航结束]



沉积物捕获器操控

序列号: 待机 电池: 2019-03-08 11:42:50 UTC 设备日期/时间 协调世界时: 2019-03-08 11:42:50 UTC

设置采样日期 (自动 | 手动) [锁定]

要锁定已输入的数据, 请点击 锁定键。

开始采集第一样品: 2019-03-08 12:00:00 UTC [日历图标]

使用采样间隔: 1 天 0 小时

产生最后样品采集时间: 2019-03-28 12:00:00 UTC [日历图标]

开始采样:

#	时间	#	时间
1	2019-03-08 12:00:00 UTC	11	2019-03-18 12:00:00 UTC
2	2019-03-09 12:00:00 UTC	12	2019-03-19 12:00:00 UTC
3	2019-03-10 12:00:00 UTC	13	2019-03-20 12:00:00 UTC
4	2019-03-11 12:00:00 UTC	14	2019-03-21 12:00:00 UTC
5	2019-03-12 12:00:00 UTC	15	2019-03-22 12:00:00 UTC
6	2019-03-13 12:00:00 UTC	16	2019-03-23 12:00:00 UTC
7	2019-03-14 12:00:00 UTC	17	2019-03-24 12:00:00 UTC
8	2019-03-15 12:00:00 UTC	18	2019-03-25 12:00:00 UTC
9	2019-03-16 12:00:00 UTC	19	2019-03-26 12:00:00 UTC
10	2019-03-17 12:00:00 UTC	20	2019-03-27 12:00:00 UTC
		End	2019-03-28 12:00:00 UTC

设置项目信息 [锁定]

要锁定已输入的数据, 请点击 锁定键。

Cruise: Yellow Sea YS_A03
Operator: Arne

项目信息

将您的项目信息输入本台沉积物捕获器。这可以包含实验, 工作站点号等。信息输入结束后, 请

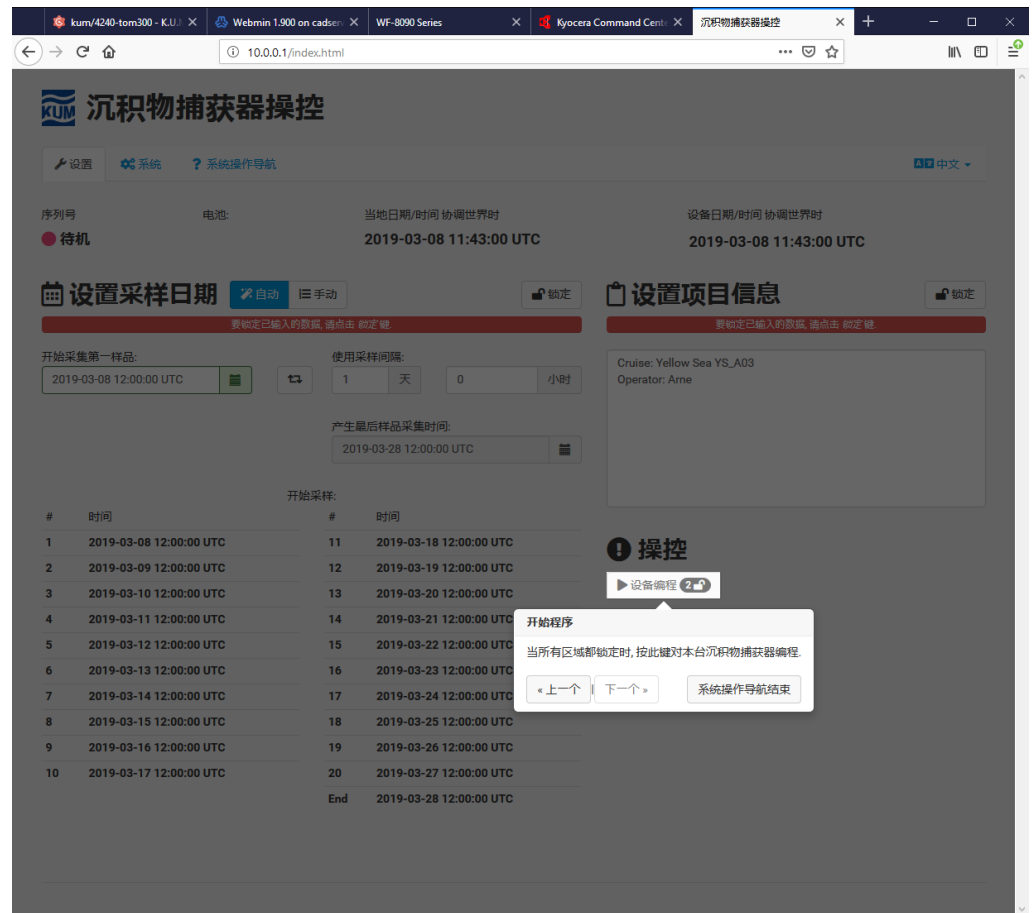
[锁定本区域]

« 上一个 | 下一个 » [系统操作导航结束]

操控

▶ 设备编程 [设备图标]

保存设置 载入设置



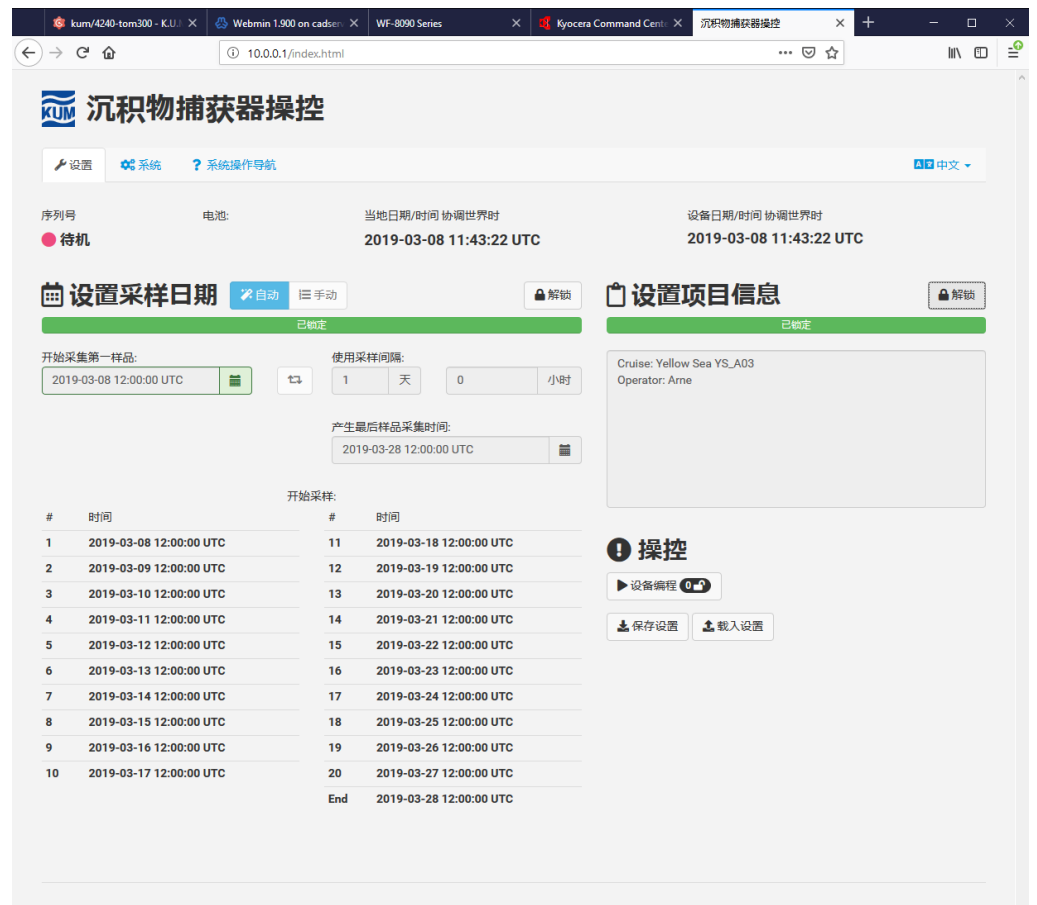
The screenshot shows the KUM Sediment Collector Control Interface in a locked state. The browser address bar displays "10.0.0.1/index.html". The interface includes a header with the KUM logo and title "沉积物捕获器操控". Below the header, there are navigation tabs for "设置" (Settings), "系统" (System), and "系统操作导航" (System Operation Navigation). The main content area displays the following information:

- 序列号 (Serial Number):** 待机 (Standby)
- 电池 (Battery):** 2019-03-08 11:43:00 UTC
- 设备日期/时间 协调世界时 (Device Date/Time Coordinated Universal Time):** 2019-03-08 11:43:00 UTC

There are two main sections for configuration:

- 设置采样日期 (Set Sampling Date):** Includes buttons for "自动" (Automatic) and "手动" (Manual). The "开始采集第一样品" (Start Collecting First Sample) is set to "2019-03-08 12:00:00 UTC". The "使用采样间隔" (Use Sampling Interval) is set to "1" day and "0" hours. The "产生最后样品采集时间" (Generate Last Sample Collection Time) is set to "2019-03-28 12:00:00 UTC".
- 设置项目信息 (Set Project Information):** Includes fields for "Cruise: Yellow Sea YS_A03" and "Operator: Arne".

A table lists the sampling schedule from 1 to 20, with times ranging from 2019-03-08 12:00:00 UTC to 2019-03-28 12:00:00 UTC. A "操控" (Control) section on the right shows a "设备编程" (Device Programming) button and a "开始程序" (Start Program) button. A tooltip indicates that the program should be started when all regions are locked.



The screenshot shows the KUM Sediment Collector Control Interface in an unlocked state. The browser address bar displays "10.0.0.1/index.html". The interface is similar to the previous one, but with the following changes:

- 序列号 (Serial Number):** 待机 (Standby)
- 电池 (Battery):** 2019-03-08 11:43:22 UTC
- 设备日期/时间 协调世界时 (Device Date/Time Coordinated Universal Time):** 2019-03-08 11:43:22 UTC

The configuration sections are now unlocked, indicated by the "已锁定" (Already Locked) status and the presence of "解锁" (Unlock) buttons:

- 设置采样日期 (Set Sampling Date):** The "开始采集第一样品" (Start Collecting First Sample) is set to "2019-03-08 12:00:00 UTC". The "使用采样间隔" (Use Sampling Interval) is set to "1" day and "0" hours. The "产生最后样品采集时间" (Generate Last Sample Collection Time) is set to "2019-03-28 12:00:00 UTC".
- 设置项目信息 (Set Project Information):** Includes fields for "Cruise: Yellow Sea YS_A03" and "Operator: Arne".

The sampling schedule table remains the same. The "操控" (Control) section now includes "保存设置" (Save Settings) and "载入设置" (Load Settings) buttons.



kum/4240-tom300 - K.U.M. Webmin 1.900 on cadser WF-8090 Series Kyocera Command Center 沉积物捕获器操控

10.0.0.1/system.html

沉积物捕获器操控

设置

系统

系统操作导航

中文

无线网络

无线网络名称
4TUNA 00000000
密码
seiamics
保存

传感器测量频率
记录倾斜仪数据每
1天

倾斜仪
校正于:
倾斜角度:
警告: 倾斜仪必须在一个水平地面校正.
我明白了
校正

数据 & 日志

下载系统日志

手动测试
警告: 采样转盘必须浸没在水中测试. 否则会导致系统永久损坏.
我明白了
采样转盘: 移动一个采样瓶

kum/4240-tom300 - K.U.M. Webmin 1.900 on cadser WF-8090 Series Kyocera Command Center 沉积物捕获器操控

10.0.0.1/system.html

沉积物捕获器操控

设置

系统

系统操作导航

中文

无线网络

无线网络名称
4TUNA 00000000
密码
seiamics
保存

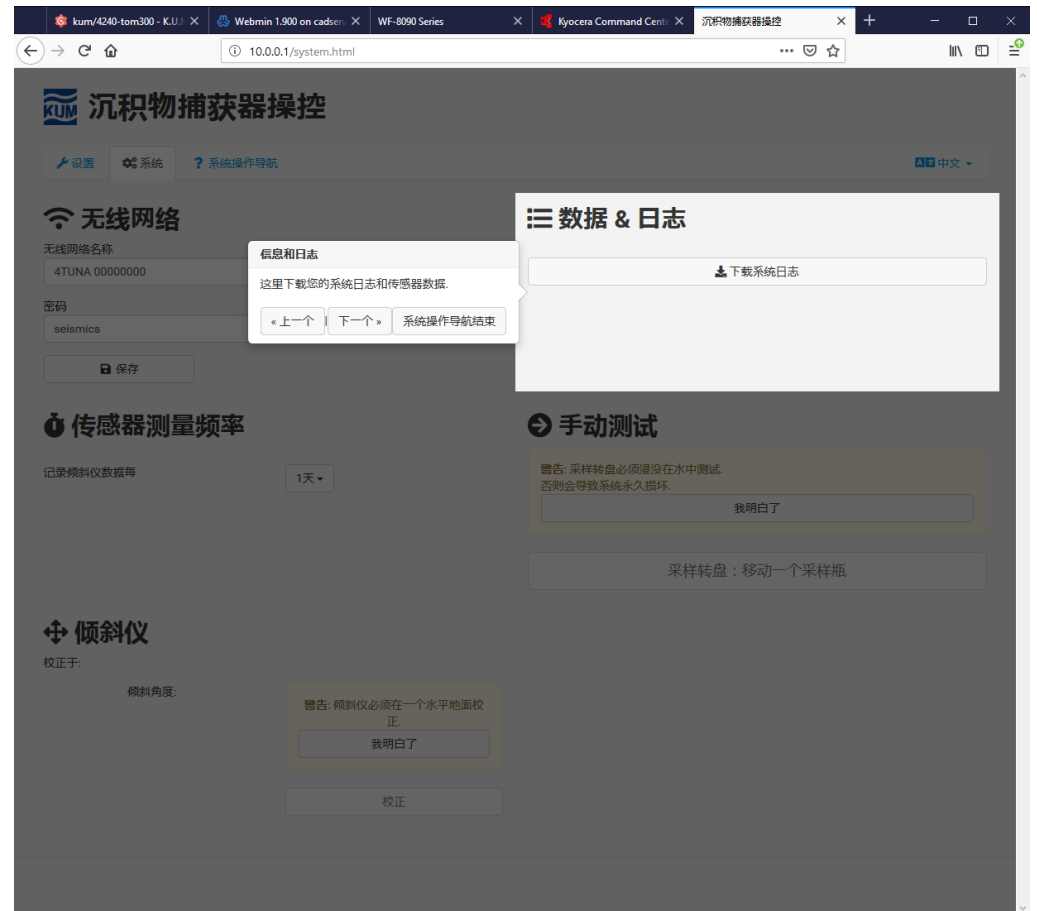
传感器测量频率
记录倾斜仪数据每
1天

倾斜仪
校正于:
倾斜角度:
警告: 倾斜仪必须在一个水平地面校正.
我明白了
校正

数据 & 日志

无线网络
这里设置您的无线网络名称和密码. 按 保存 键来重新启动带有新设置的无线网络.
« 上一个 | 下一个 » 系统操作导航结束

手动测试
警告: 采样转盘必须浸没在水中测试. 否则会导致系统永久损坏.
我明白了
采样转盘: 移动一个采样瓶



沉积物捕获器操控

设置 系统 系统操作导航

中文

无线网络

无线网络名称: 4TUNA 00000000

密码: seismics

保存

传感器测量频率

记录倾斜仪数据每: 1天

倾斜仪

校正于:

倾斜角度:

警告: 倾斜仪必须在一个水平地面校正。

我明白了

校正

数据 & 日志

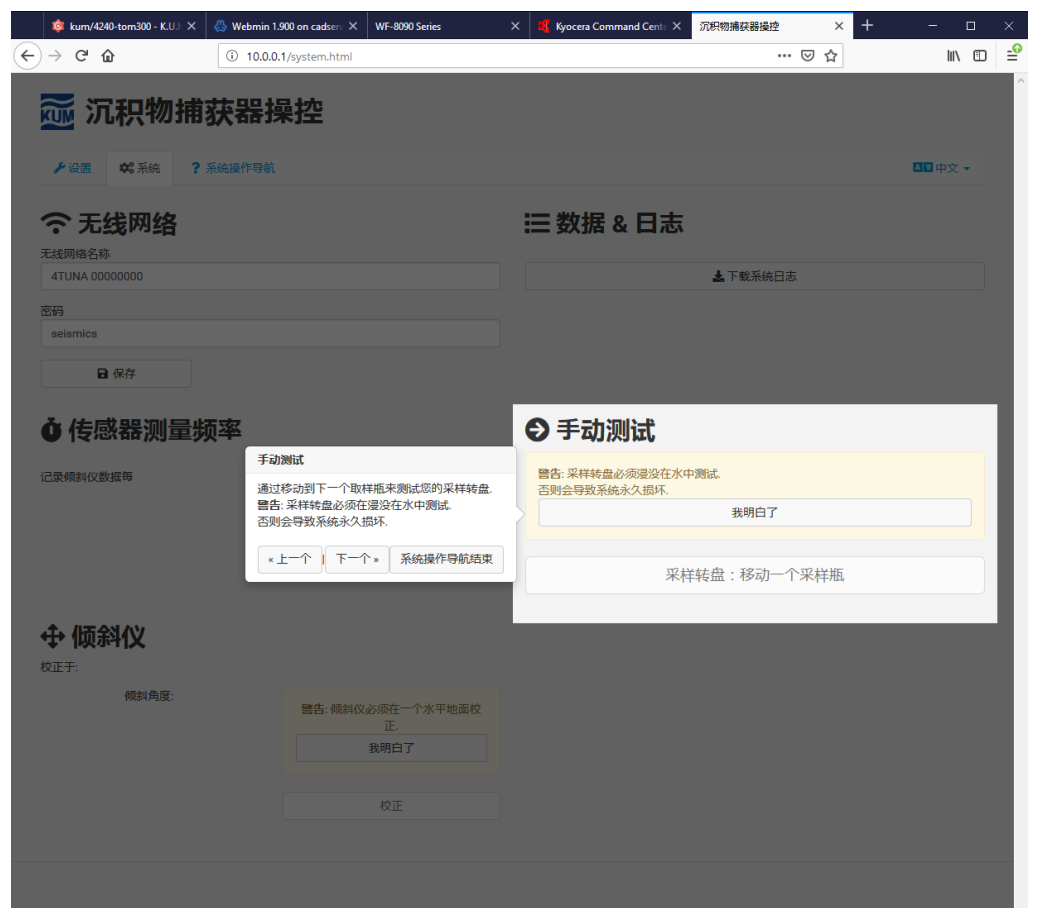
下载系统日志

手动测试

警告: 采样转盘必须浸没在水中测试。否则会导致系统永久损坏。

我明白了

采样转盘: 移动一个采样瓶



沉积物捕获器操控

设置 系统 系统操作导航

中文

无线网络

无线网络名称: 4TUNA 00000000

密码: seismics

保存

传感器测量频率

记录倾斜仪数据每

倾斜仪

校正于:

倾斜角度:

警告: 倾斜仪必须在一个水平地面校正。

我明白了

校正

数据 & 日志

下载系统日志

手动测试

警告: 采样转盘必须浸没在水中测试。否则会导致系统永久损坏。

我明白了

采样转盘: 移动一个采样瓶

